



Data Analytics &
Data Engineering

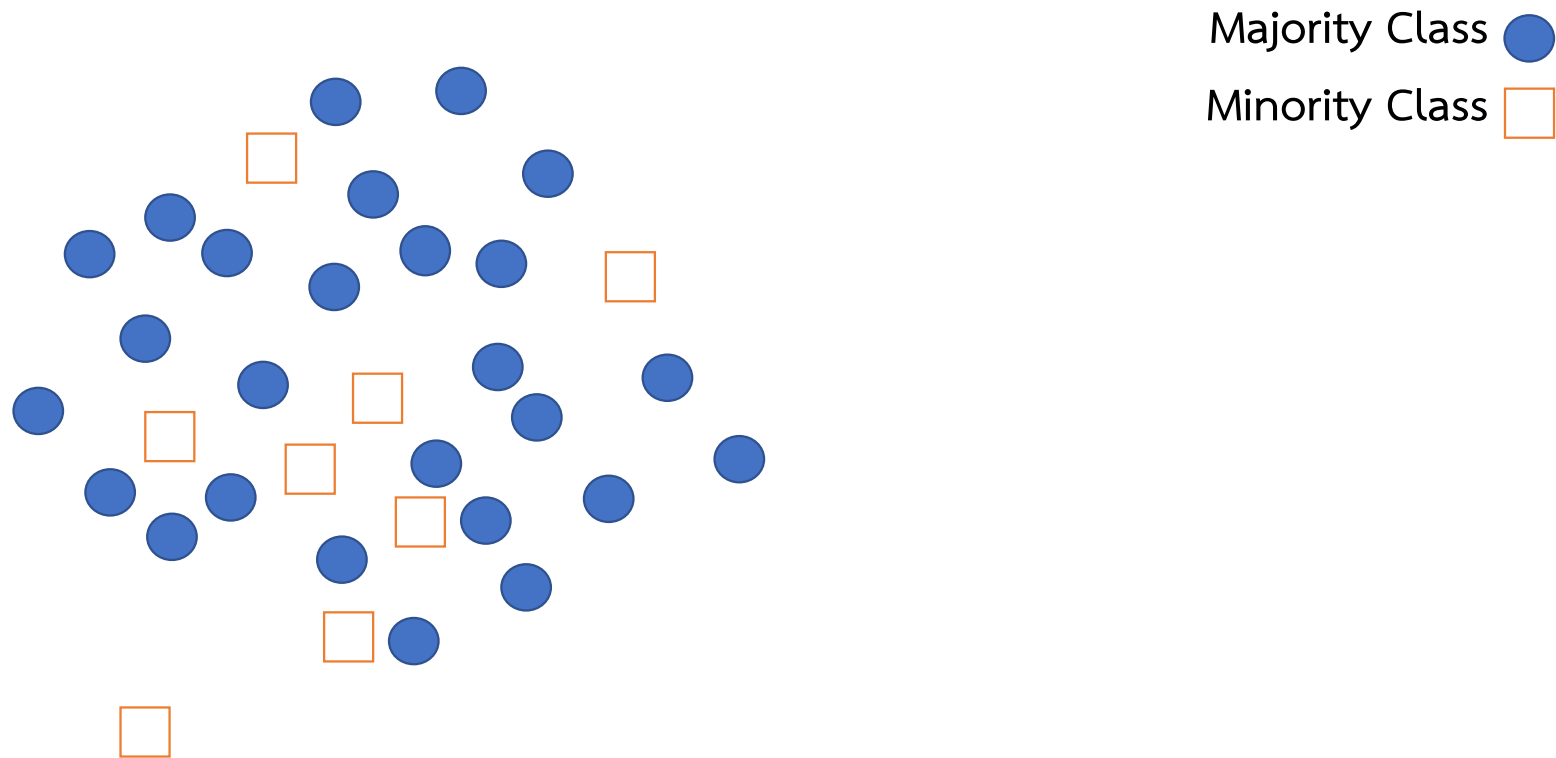
KNIME Case Study #2:
Classification Model

Damrongrit Setsirichok
Waranyu Wongseree

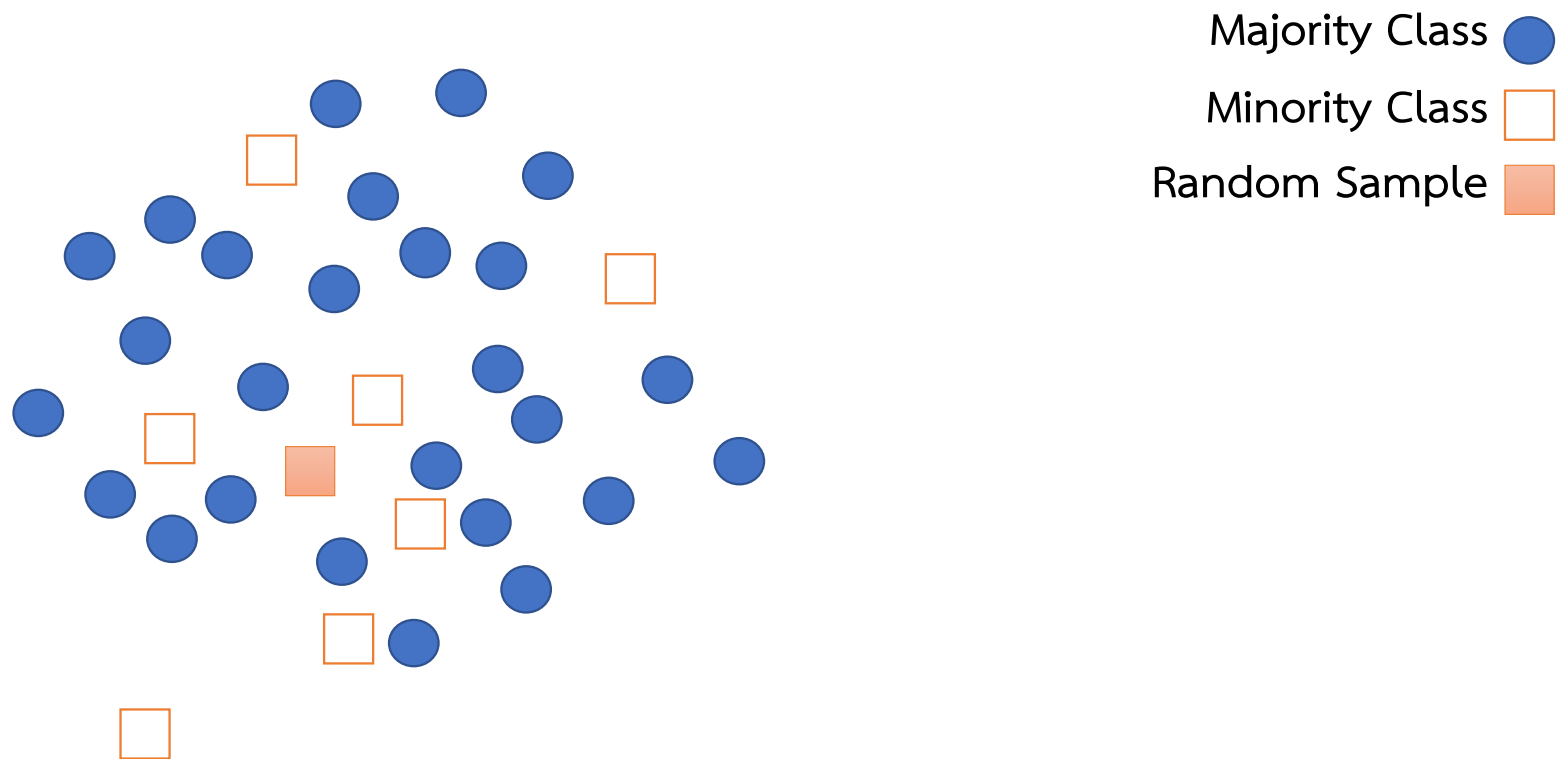
KNIME: Class Balancing



Synthetic Minority Oversampling Technique (SMOTE)

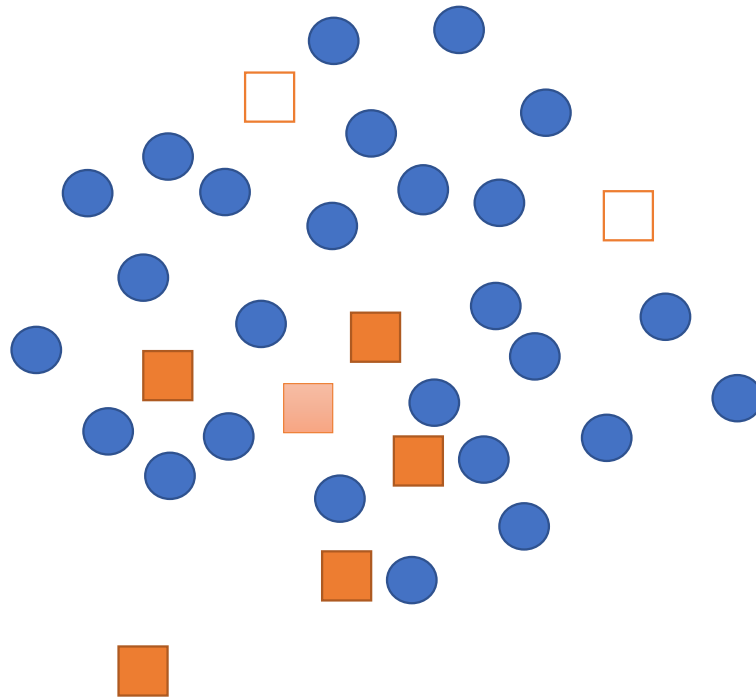


Synthetic Minority Oversampling Technique (SMOTE)



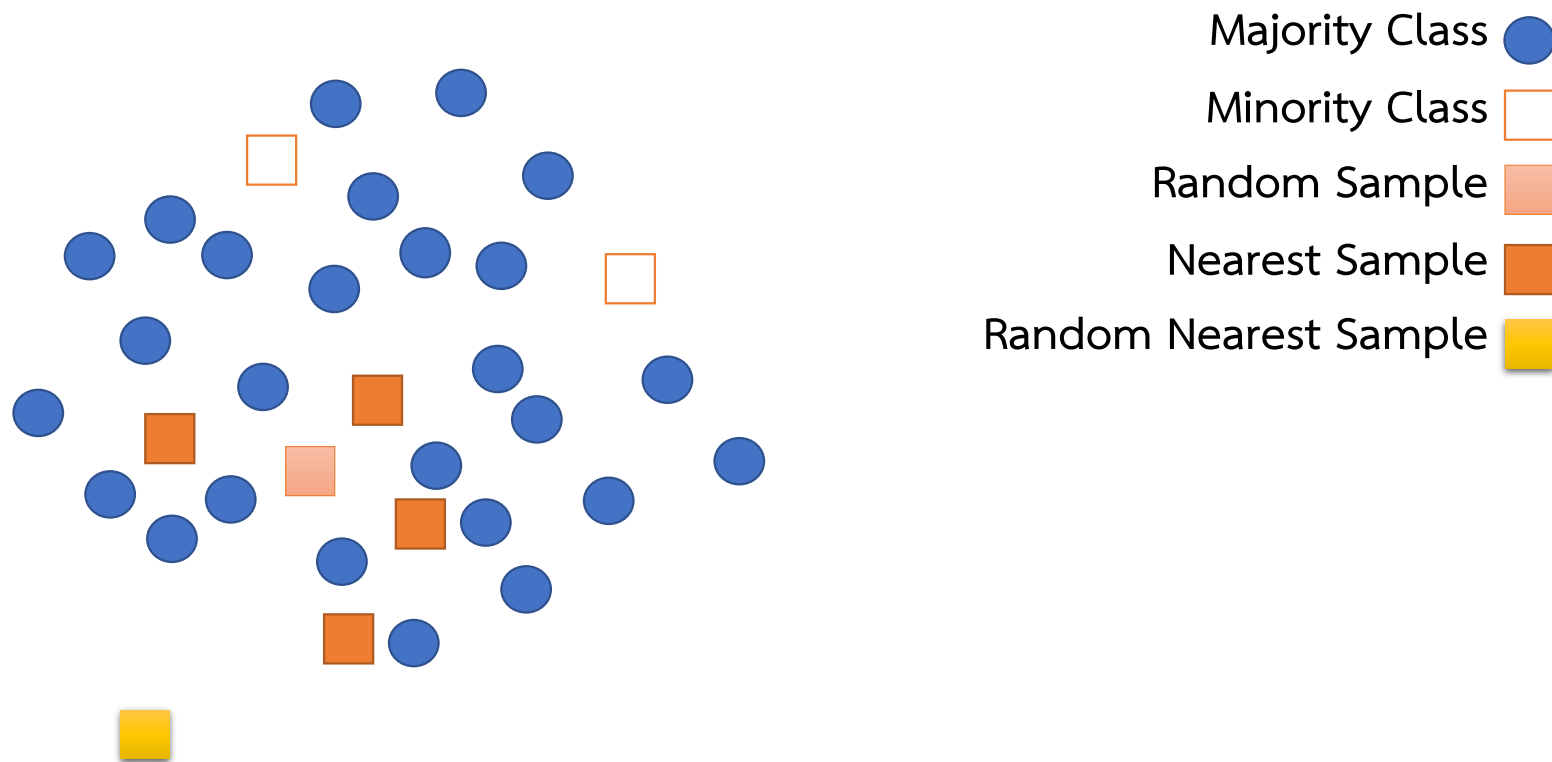
Synthetic Minority Oversampling Technique (SMOTE)

Hyper-parameter ($K = 5$)

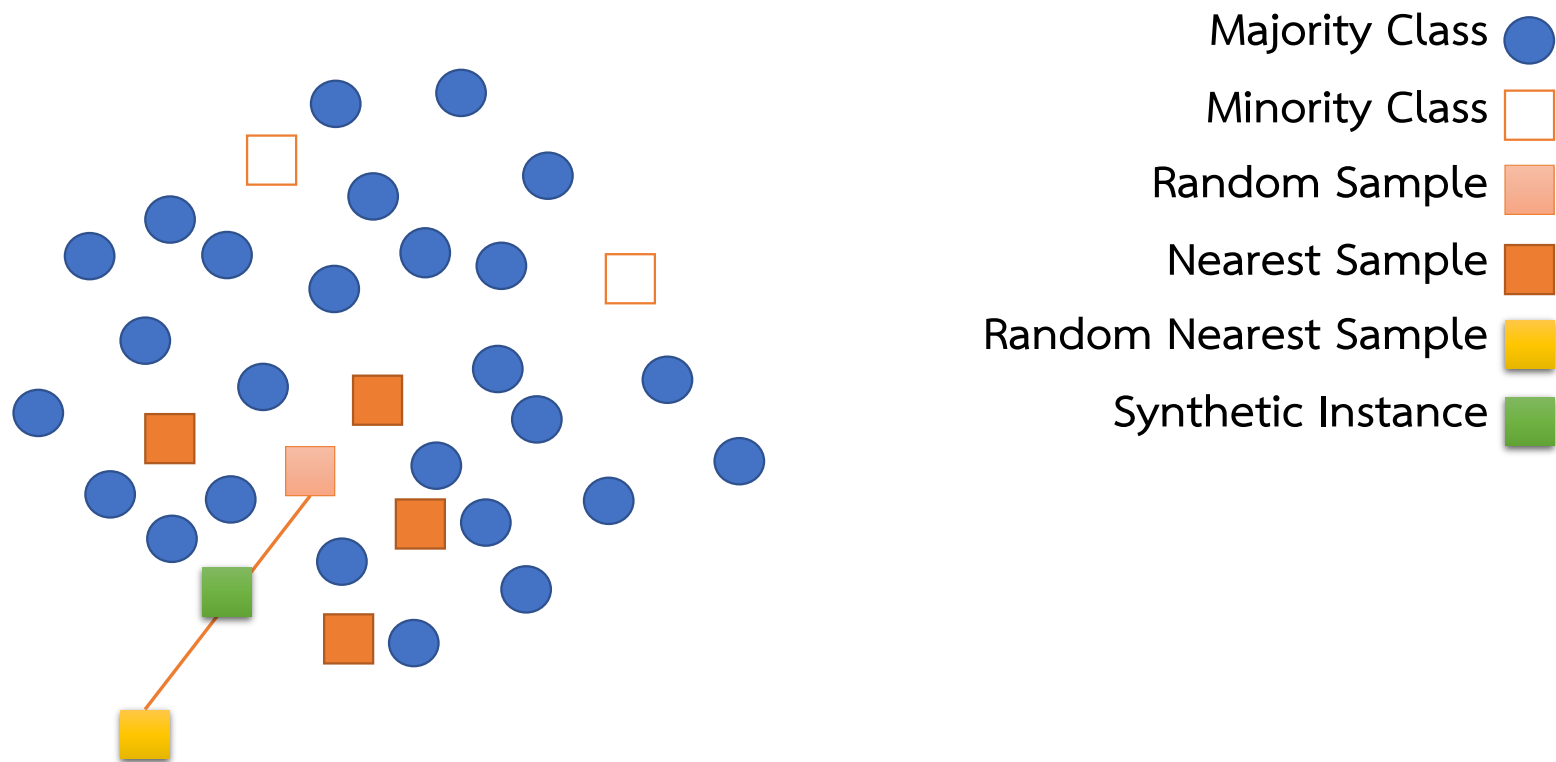


- Majority Class ●
- Minority Class □
- Random Sample ■
- Nearest Samples ■

Synthetic Minority Oversampling Technique (SMOTE)



Synthetic Minority Oversampling Technique (SMOTE)

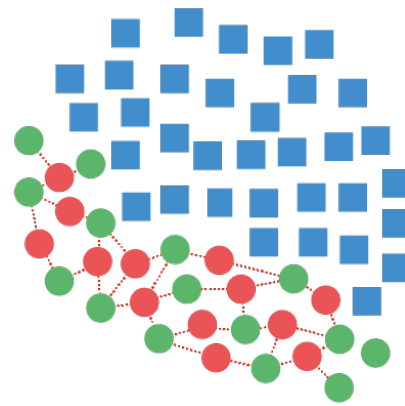


Class Balancing

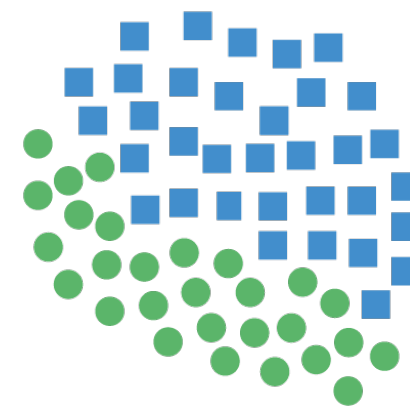
Synthetic Minority Oversampling Technique



Original Dataset

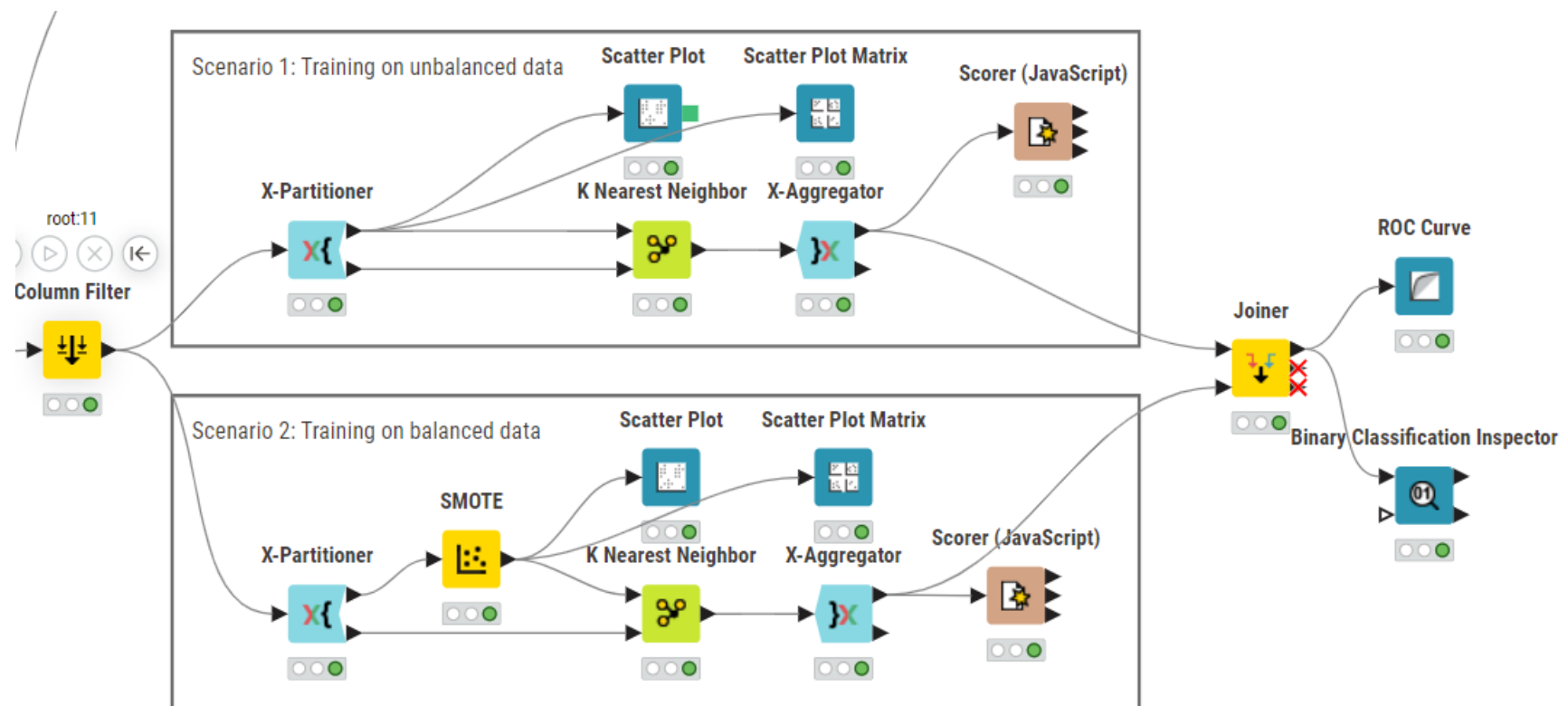


Generating Samples

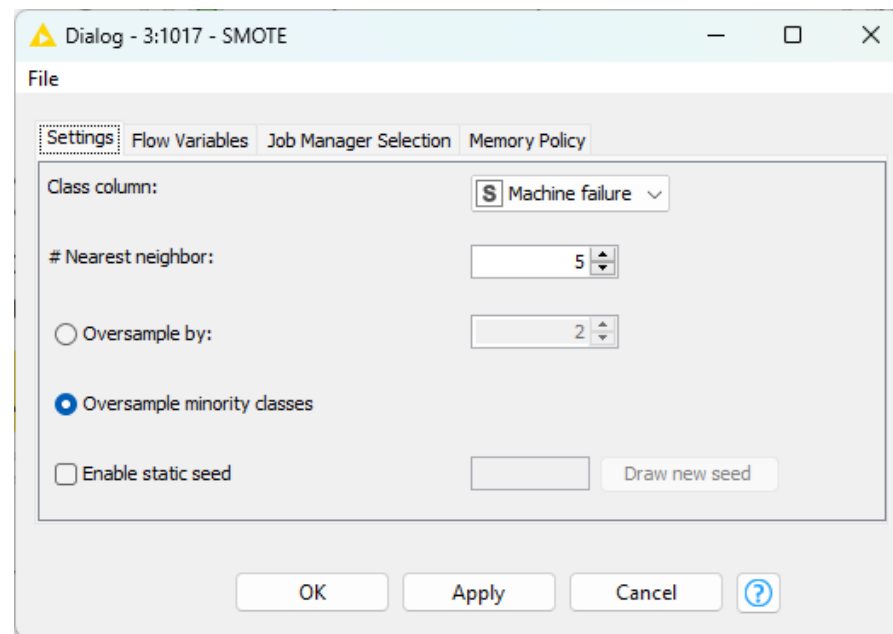


Resampled Dataset

Classifier: k Nearest Neighbor

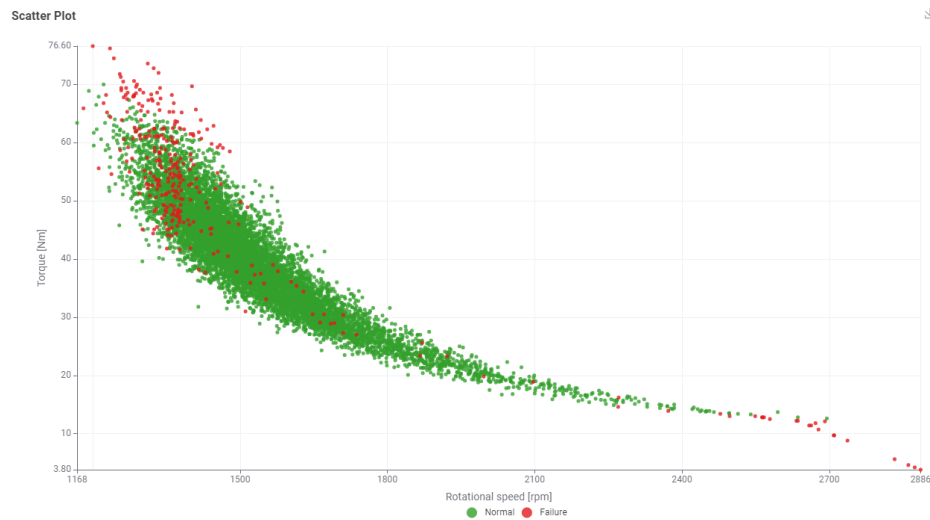


Important Setting: SMOTE

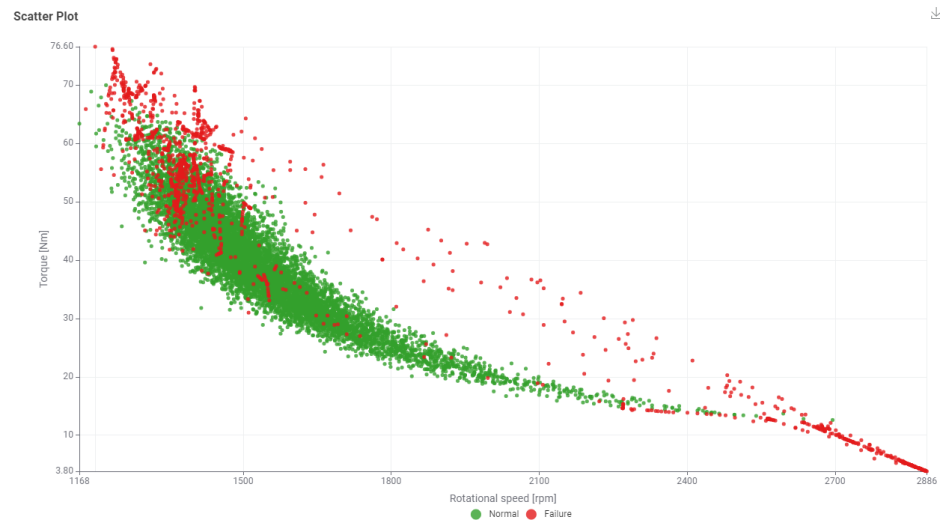


Scatter Plot

Unbalanced Data

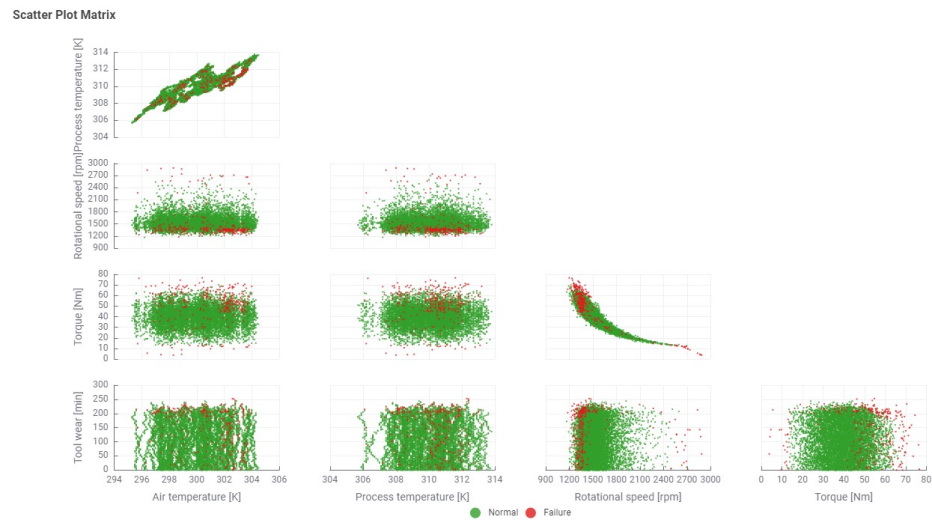


Balanced Data

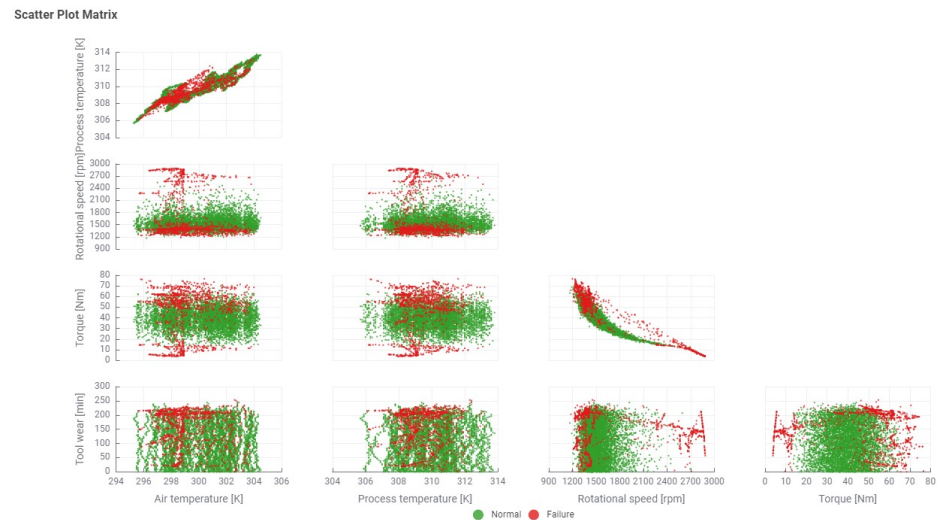


Matrix of Scatter Plot

Unbalanced Data



Balanced Data



Confusion Metrix: k Nearest Neighbor

Unbalanced Data

	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	74	265	21.83%
Normal (Actual)	66	9595	99.32%
	52.86%	97.31%	

Balanced Data

	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	211	128	62.24%
Normal (Actual)	903	8758	90.65%
	18.94%	98.56%	

Performance Metrics: k Nearest Neighbor

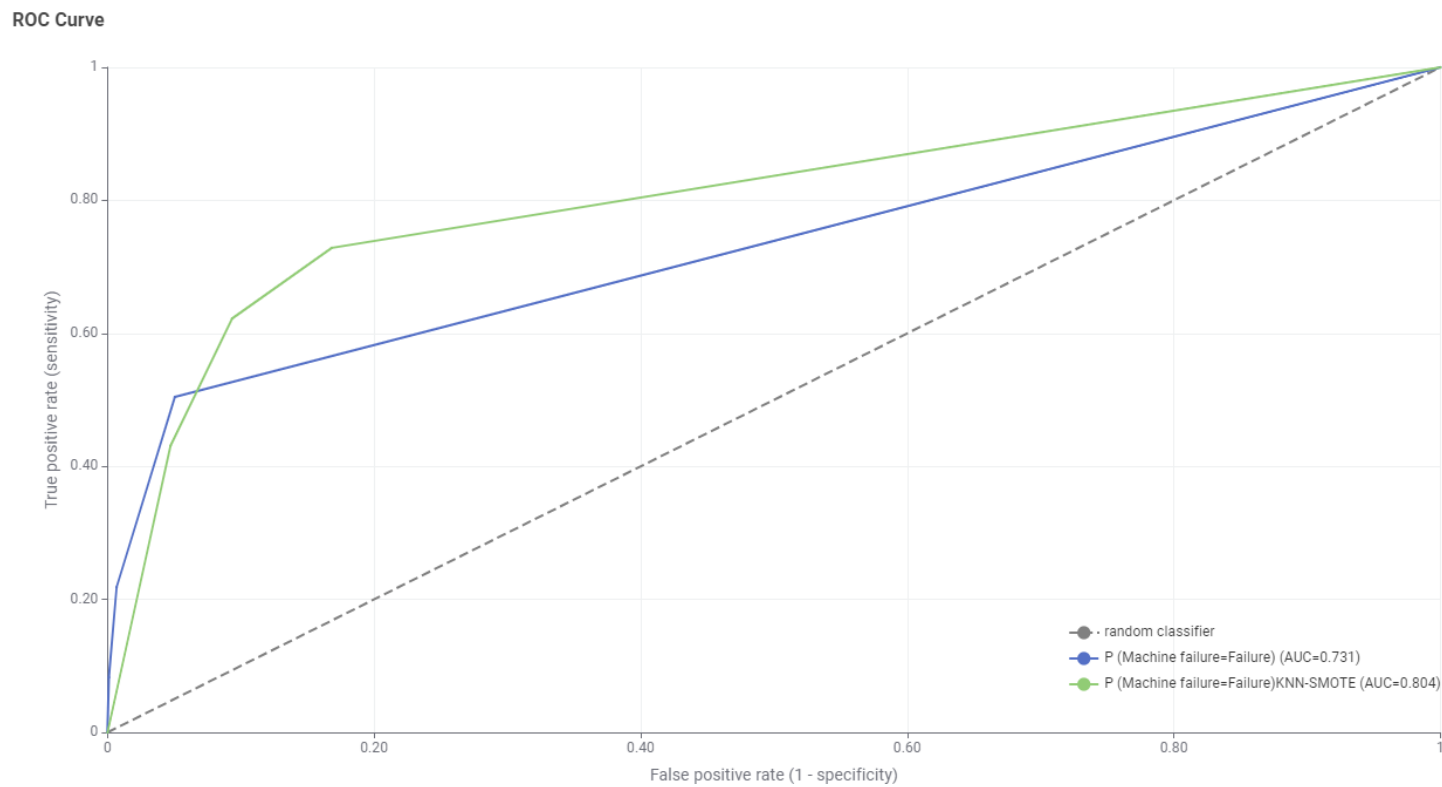
Unbalanced Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	74	66	9595	265	21.83%	52.86%	21.83%	99.32%	30.90%
Normal	9595	265	74	66	99.32%	97.31%	99.32%	21.83%	98.30%

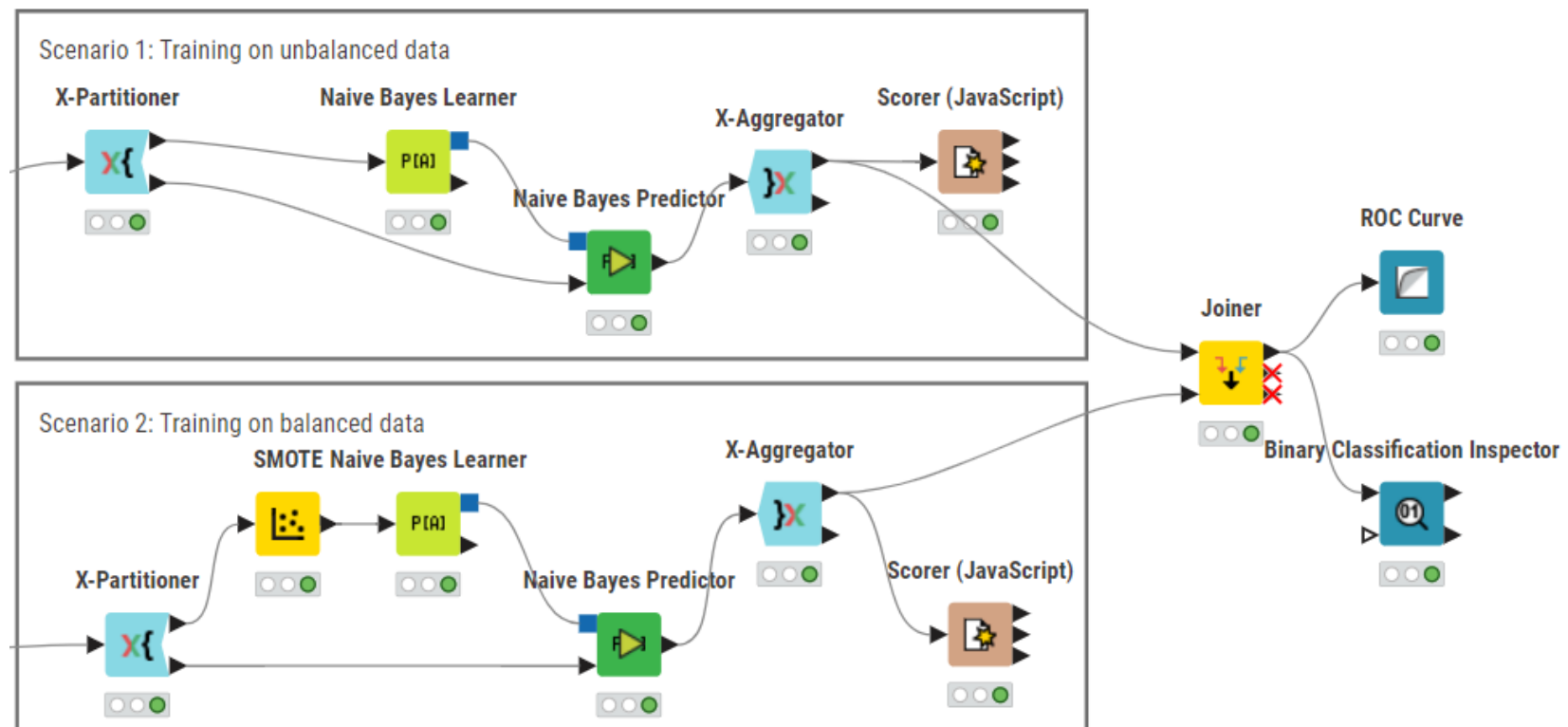
Balanced Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	211	903	8758	128	62.24%	18.94%	62.24%	90.65%	29.04%
Normal	8758	128	211	903	90.65%	98.56%	90.65%	62.24%	94.44%

ROC Curve: k Nearest Neighbor



Classifier: Naïve Bayes



Confusion Matrix: Naïve Bayes

Unbalanced Data

Confusion Matrix (10000 displayed rows)

	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	34	305	Sensitivity 0.100295
Normal (Actual)	8	9653	Specificity 0.999172
	Precision 0.809524	NPV 0.969371	

Balanced Data

Confusion Matrix (10000 displayed rows)

	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	225	114	Sensitivity 0.663717
Normal (Actual)	1179	8482	Specificity 0.877963
	Precision 0.160256	NPV 0.986738	

Performance Metrics: Naïve Bayes

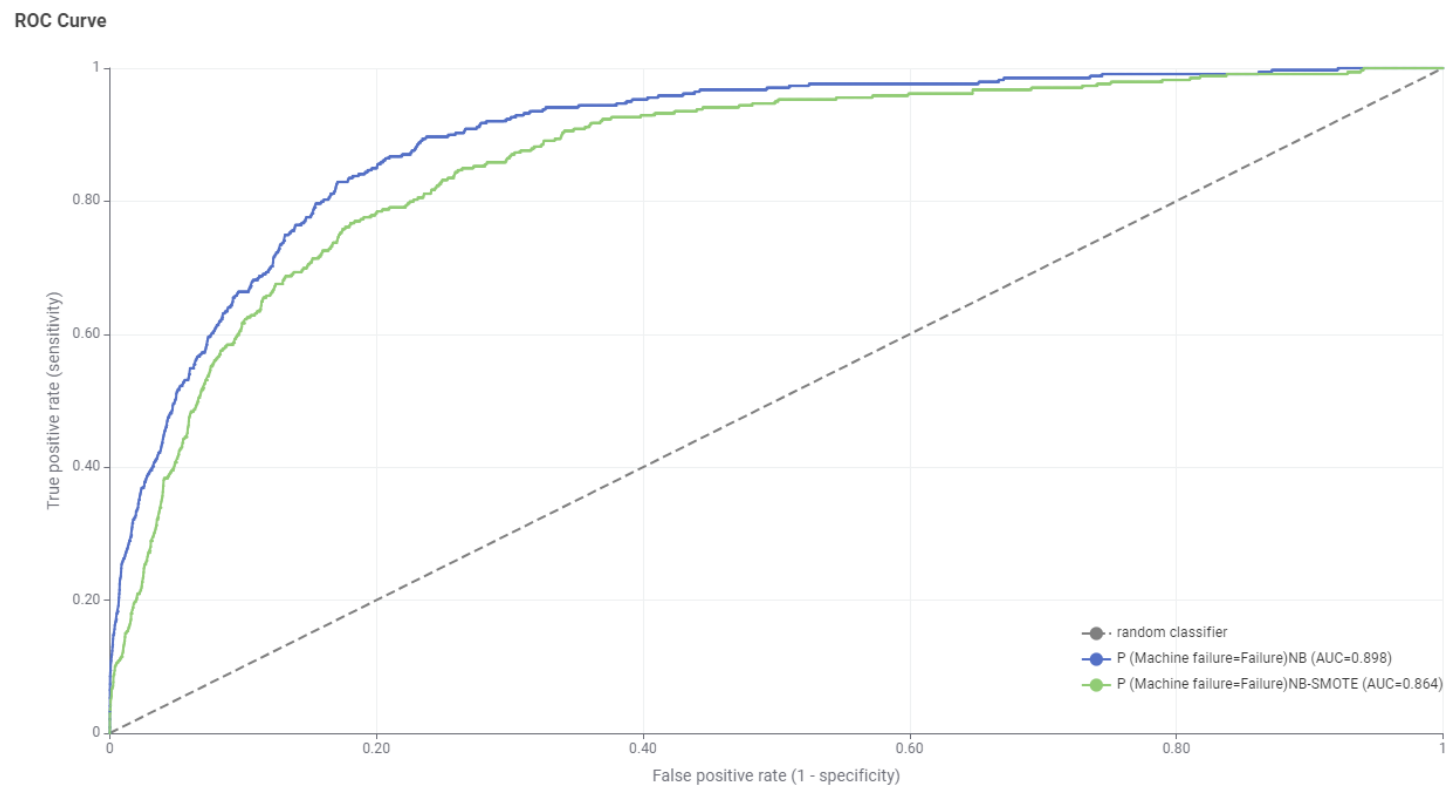
Unbalanced Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	34	8	9653	305	10.03%	80.95%	10.03%	99.92%	17.85%
Normal	9653	305	34	8	99.92%	96.94%	99.92%	10.03%	98.40%

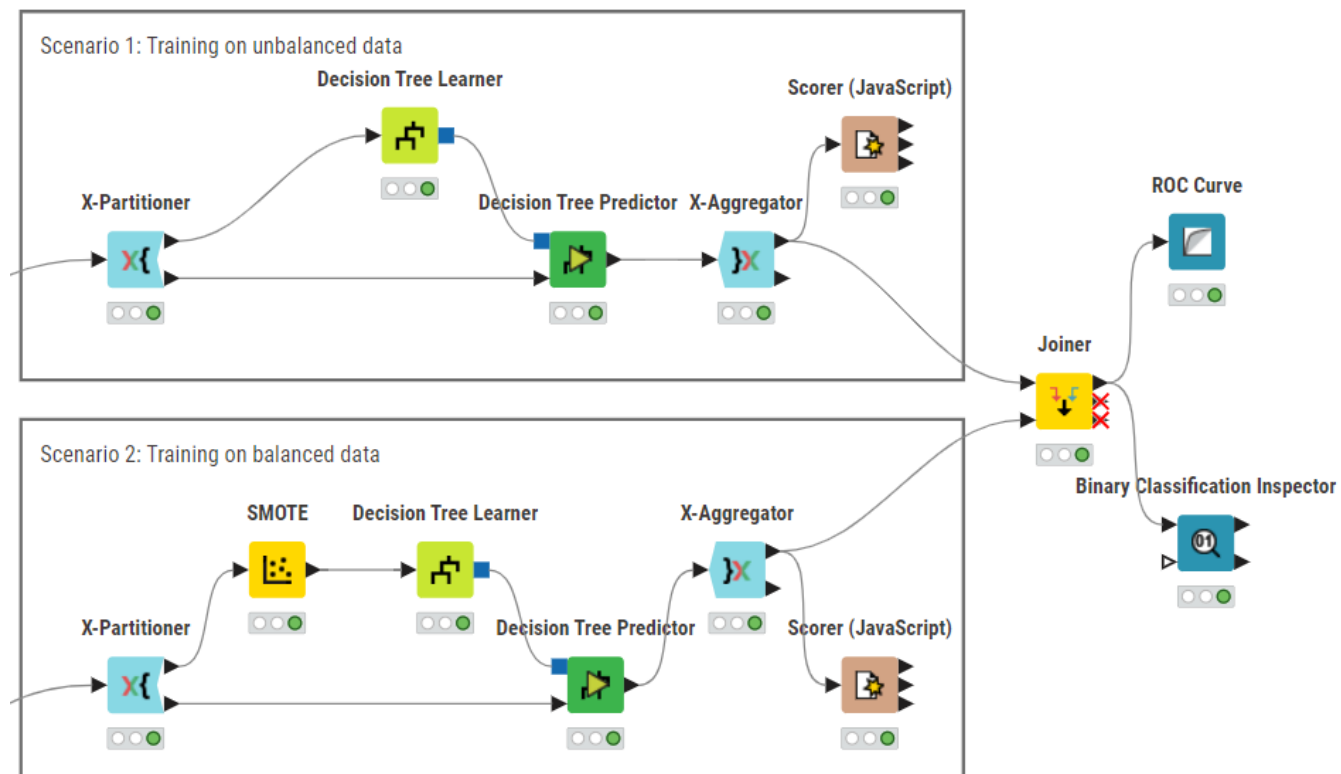
Balanced Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	225	1179	8482	114	66.37%	16.03%	66.37%	87.80%	25.82%
Normal	8482	114	225	1179	87.80%	98.67%	87.80%	66.37%	92.92%

ROC Curve: Naïve Bayes

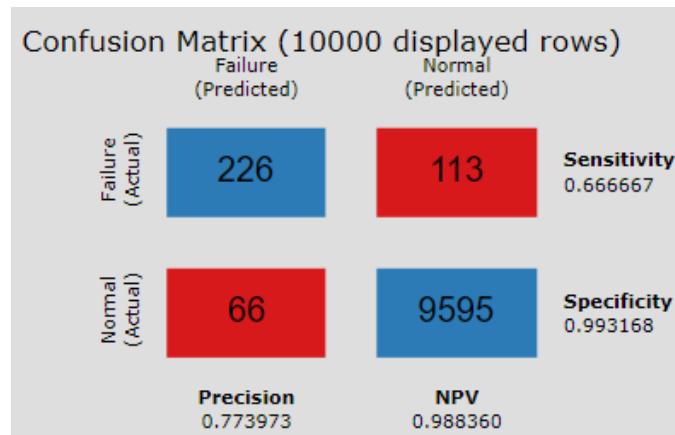


Classifier: Decision Tree

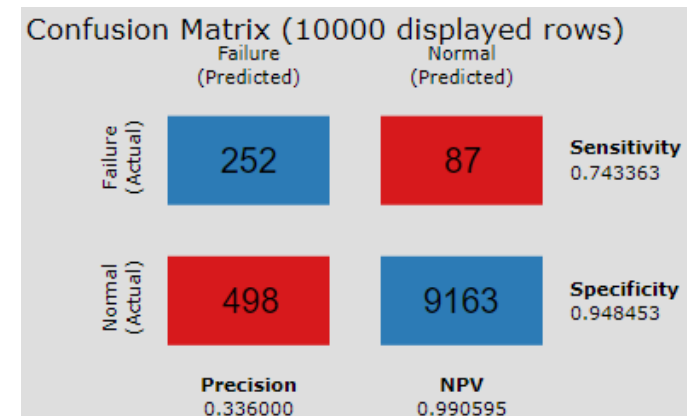


Confusion Matrix: Decision Tree

Unbalanced Data



Balanced Data



Performance Metrics: Decision Tree

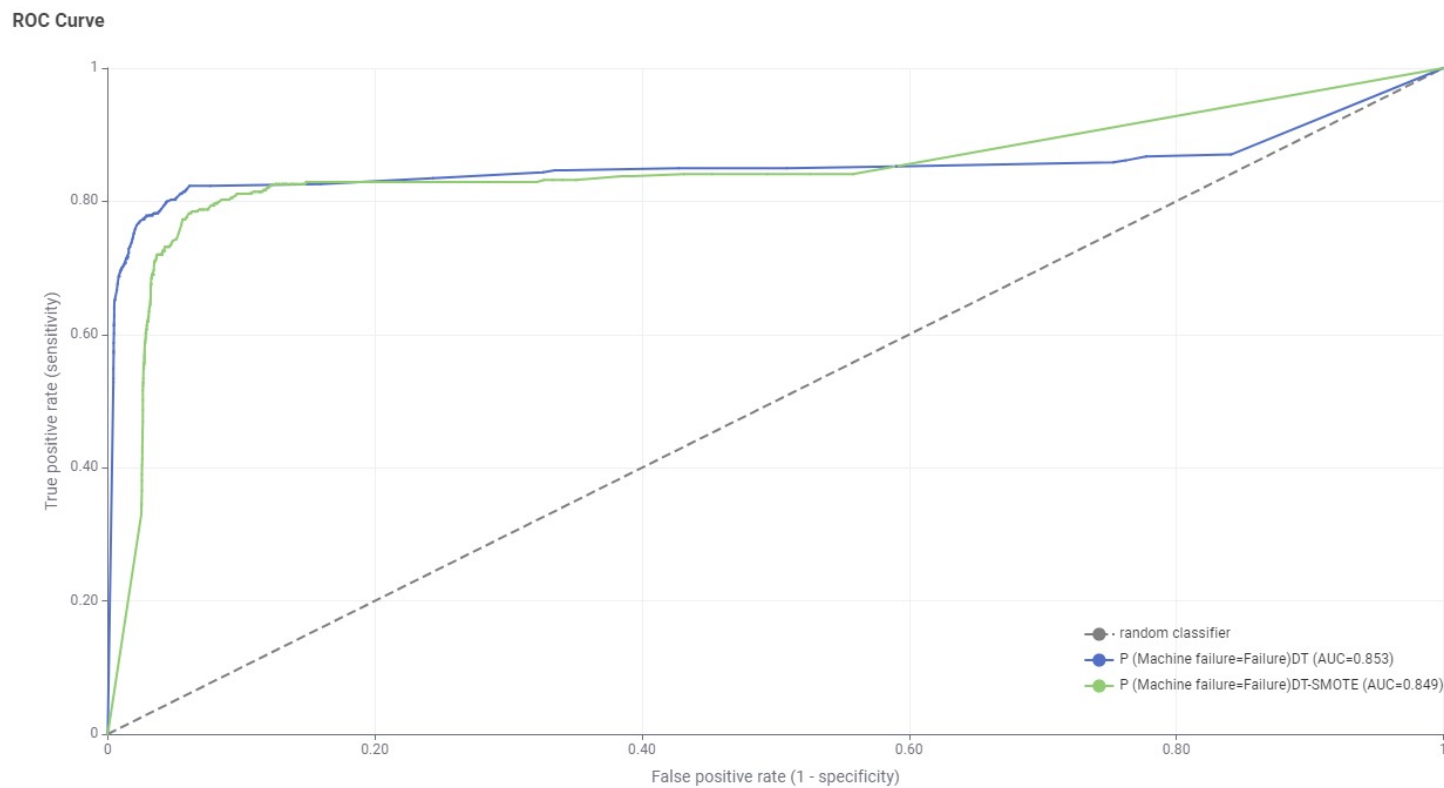
Unbalanced Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	226	66	9595	113	66.67%	77.40%	66.67%	99.32%	71.63%
Normal	9595	113	226	66	99.32%	98.84%	99.32%	66.67%	99.08%

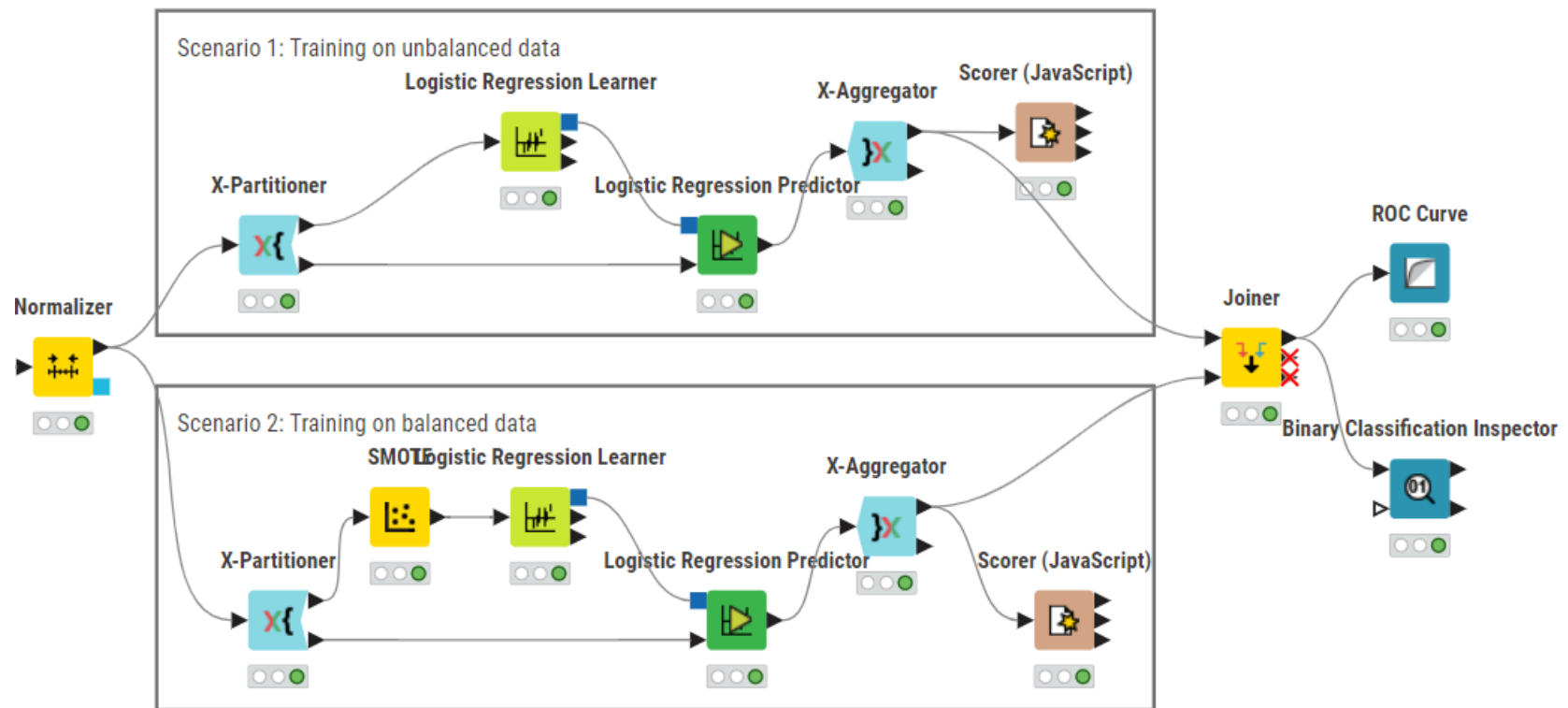
Balanced Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	252	498	9163	87	74.34%	33.60%	74.34%	94.85%	46.28%
Normal	9163	87	252	498	94.85%	99.06%	94.85%	74.34%	96.91%

ROC Curve: Decision Tree

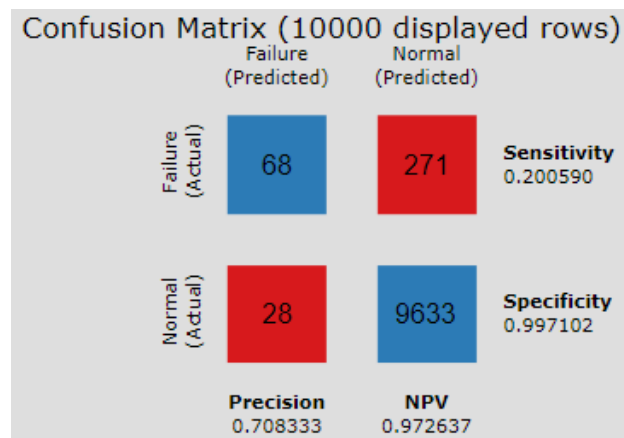


Classifier: Logistic Regression

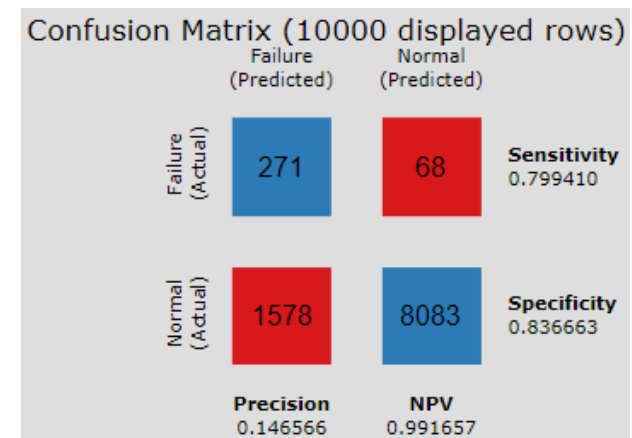


Confusion Matrix: Logistic Regression

Unbalanced Data



Balanced Data



Performance Metrics: Logistic Regression

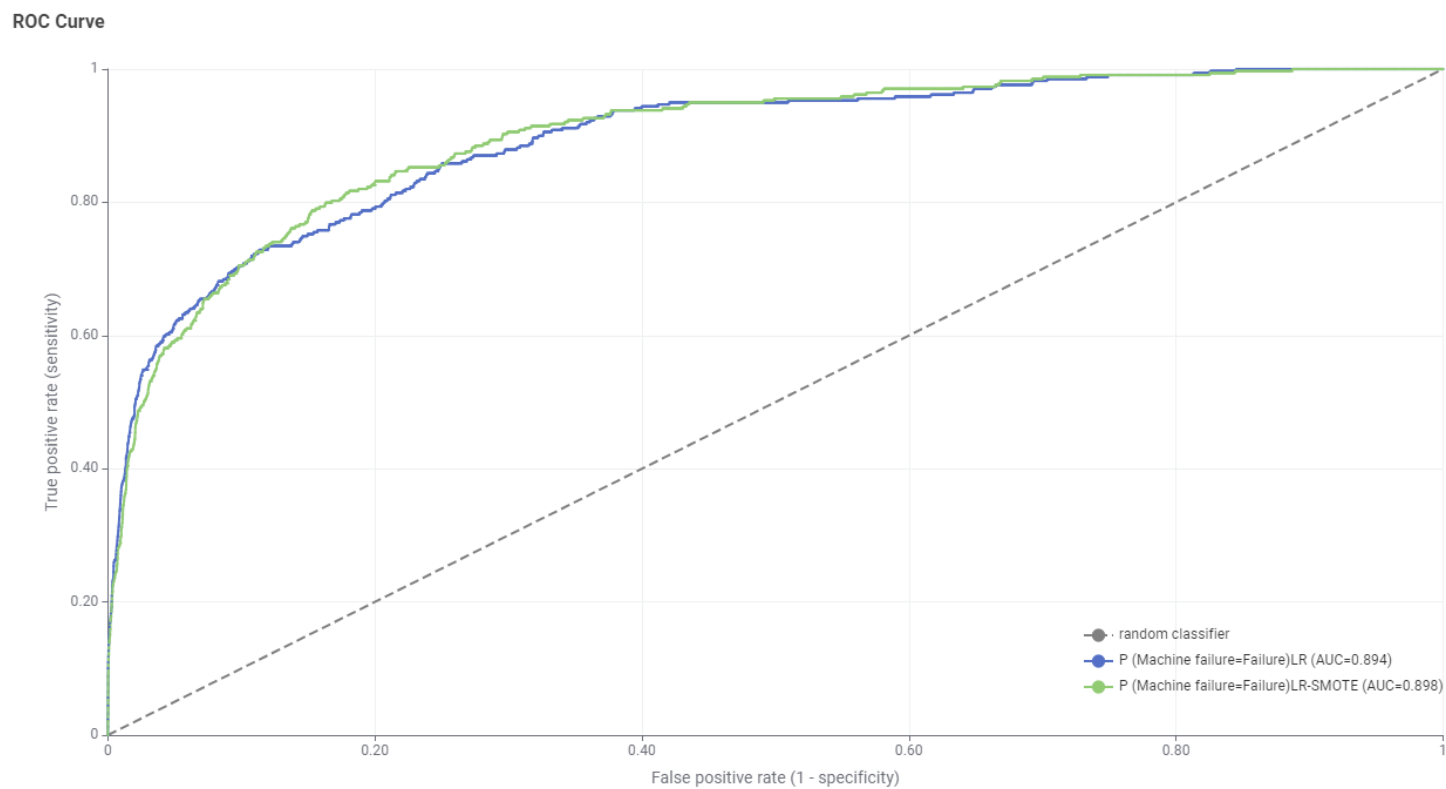
Unbalanced Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	69	28	9633	270	20.35%	71.13%	20.35%	99.71%	31.65%
Normal	9633	270	69	28	99.71%	97.27%	99.71%	20.35%	98.48%

Balanced Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	271	1579	8082	68	79.94%	14.65%	79.94%	83.66%	24.76%
Normal	8082	68	271	1579	83.66%	99.17%	83.66%	79.94%	90.75%

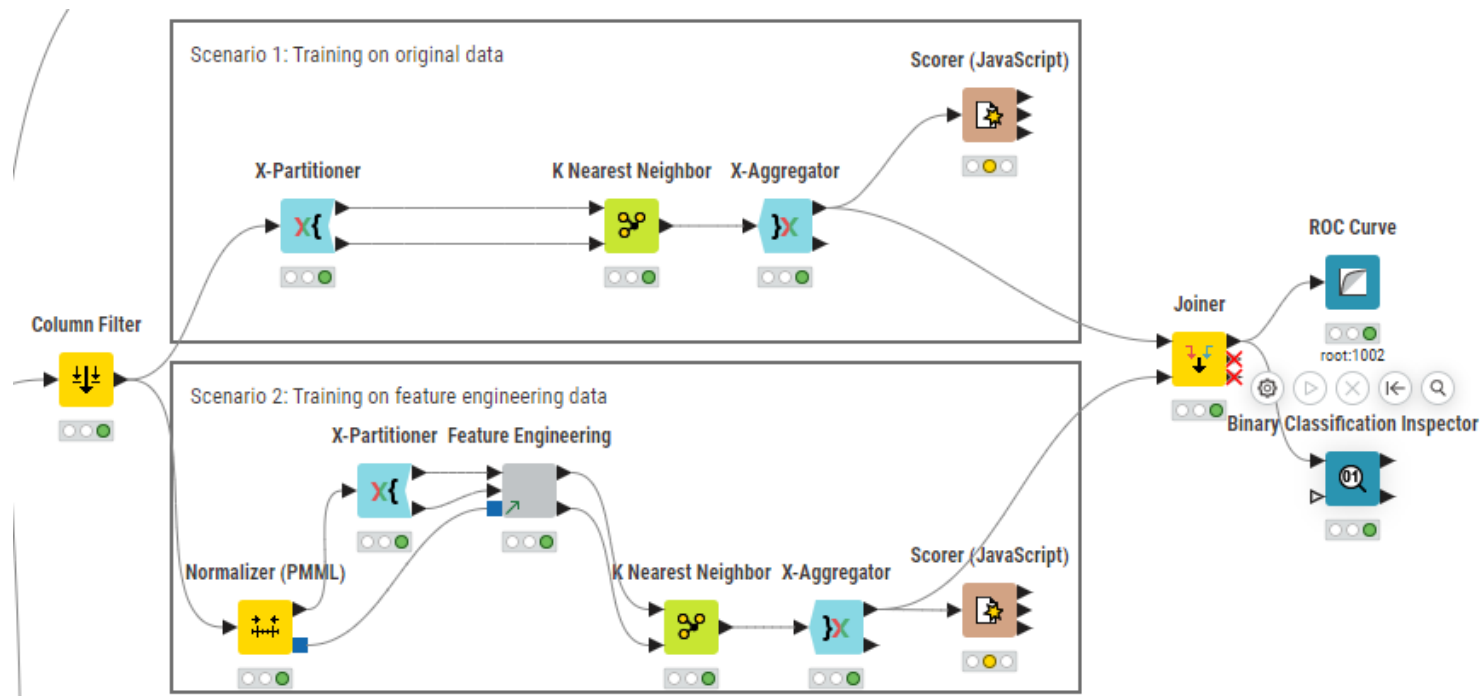
ROC Curve: Logistic Regression



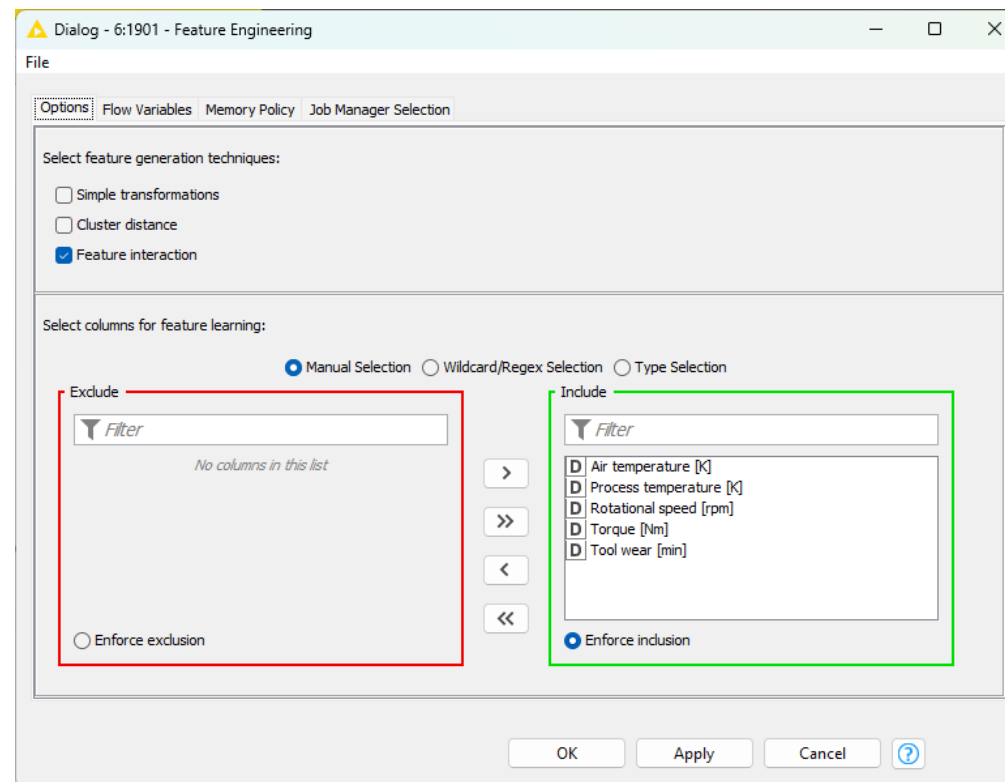
KNIME: Feature Engineering



Classifier: k Nearest Neighbor



Important Setting: Feature Engineering



Confusion Metrix: k Nearest Neighbor

Original Data

	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	74	265	21.83%
Normal (Actual)	66	9595	99.32%
	52.86%	97.31%	

Feature Engineering Data

	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	102	237	30.09%
Normal (Actual)	59	9602	99.39%
	63.35%	97.59%	

Performance Metrics: k Nearest Neighbor

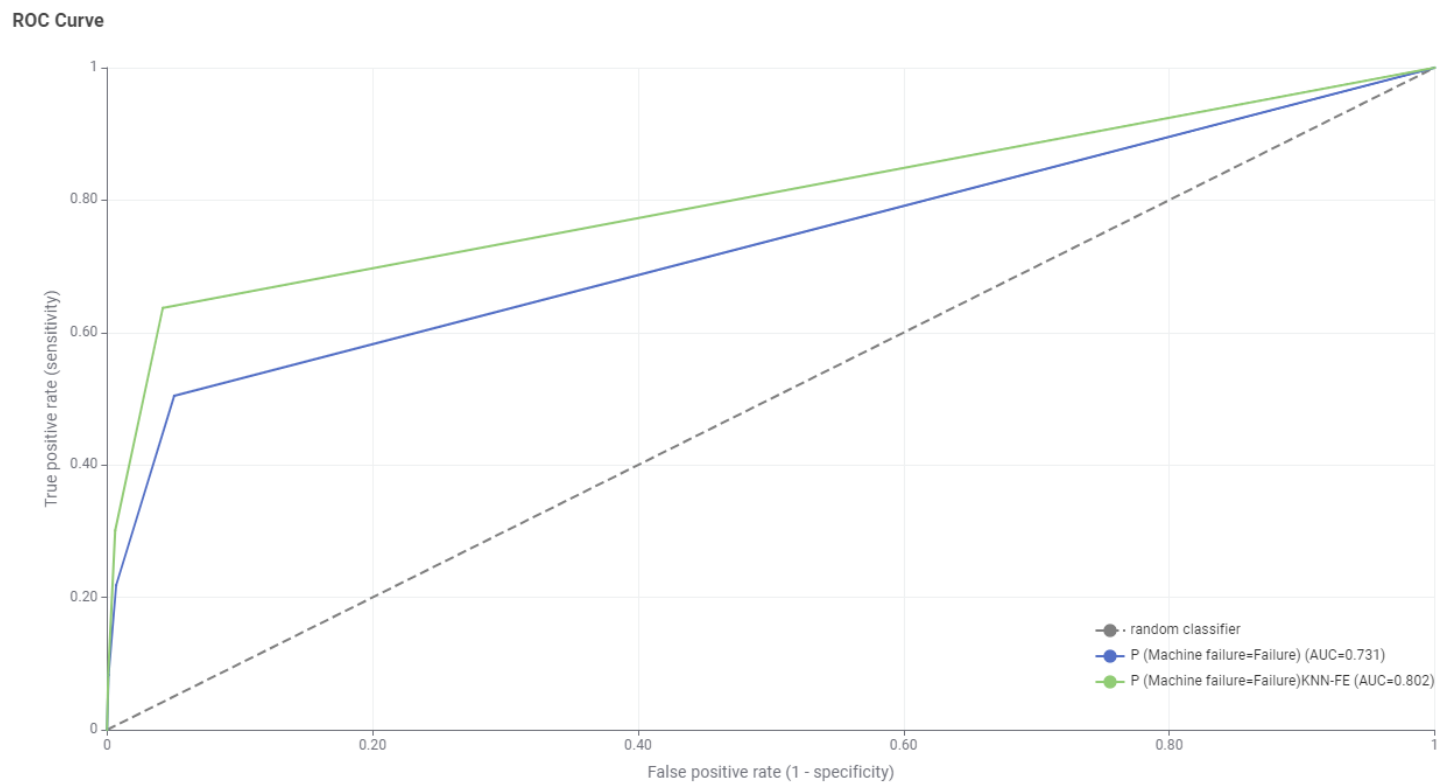
Original Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	74	66	9595	265	21.83%	52.86%	21.83%	99.32%	30.90%
Normal	9595	265	74	66	99.32%	97.31%	99.32%	21.83%	98.30%

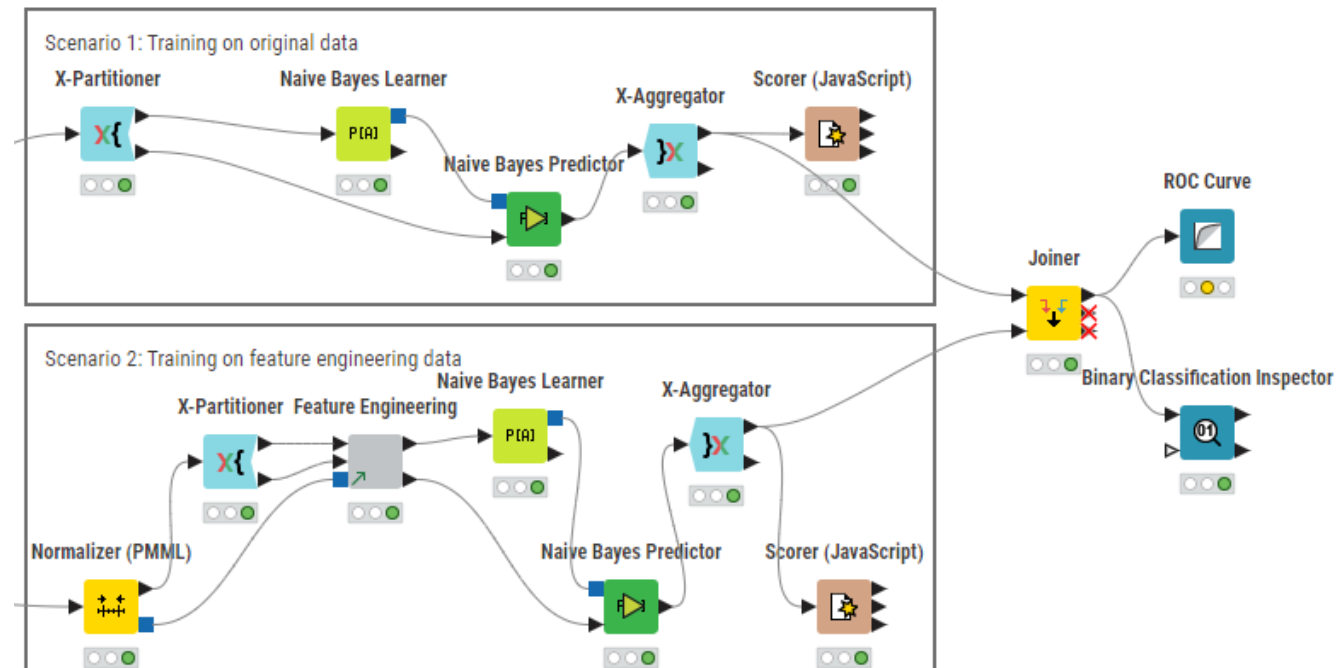
Feature Engineering Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	102	59	9602	237	30.09%	63.35%	30.09%	99.39%	40.80%
Normal	9602	237	102	59	99.39%	97.59%	99.39%	30.09%	98.48%

ROC Curve: k Nearest Neighbor



Classifier: Naïve Bayes



Confusion Matrix: Naïve Bayes

Original Data

Confusion Matrix (10000 displayed rows)

	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	34	305	Sensitivity 0.100295
Normal (Actual)	8	9653	Specificity 0.999172
	Precision 0.809524	NPV 0.969371	

Feature Engineering Data

Confusion Matrix (10000 displayed rows)

	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	324	15	Sensitivity 0.955752
Normal (Actual)	7098	2563	Specificity 0.265293
	Precision 0.043654	NPV 0.994182	

Performance Metrics: Naïve Bayes

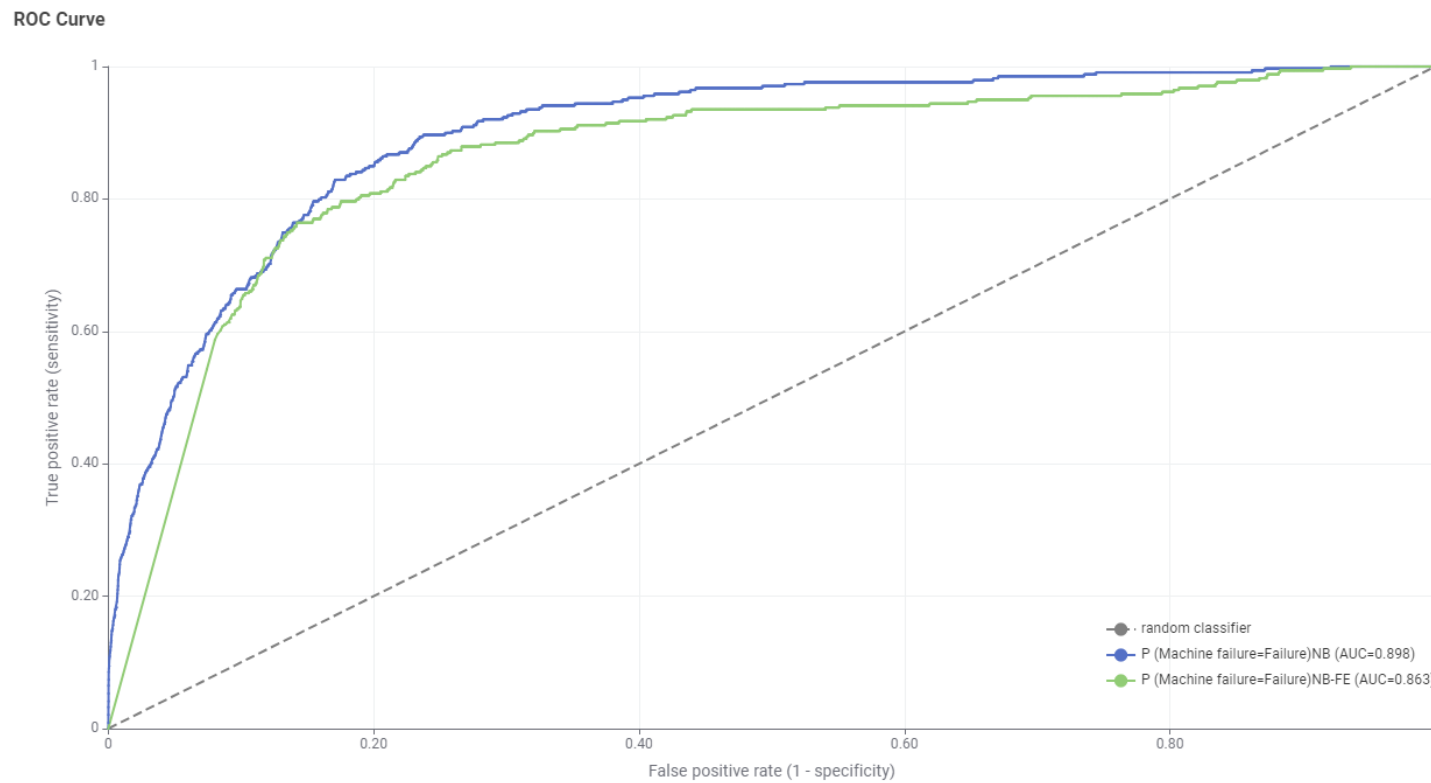
Original Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	34	8	9653	305	10.03%	80.95%	10.03%	99.92%	17.85%
Normal	9653	305	34	8	99.92%	96.94%	99.92%	10.03%	98.40%

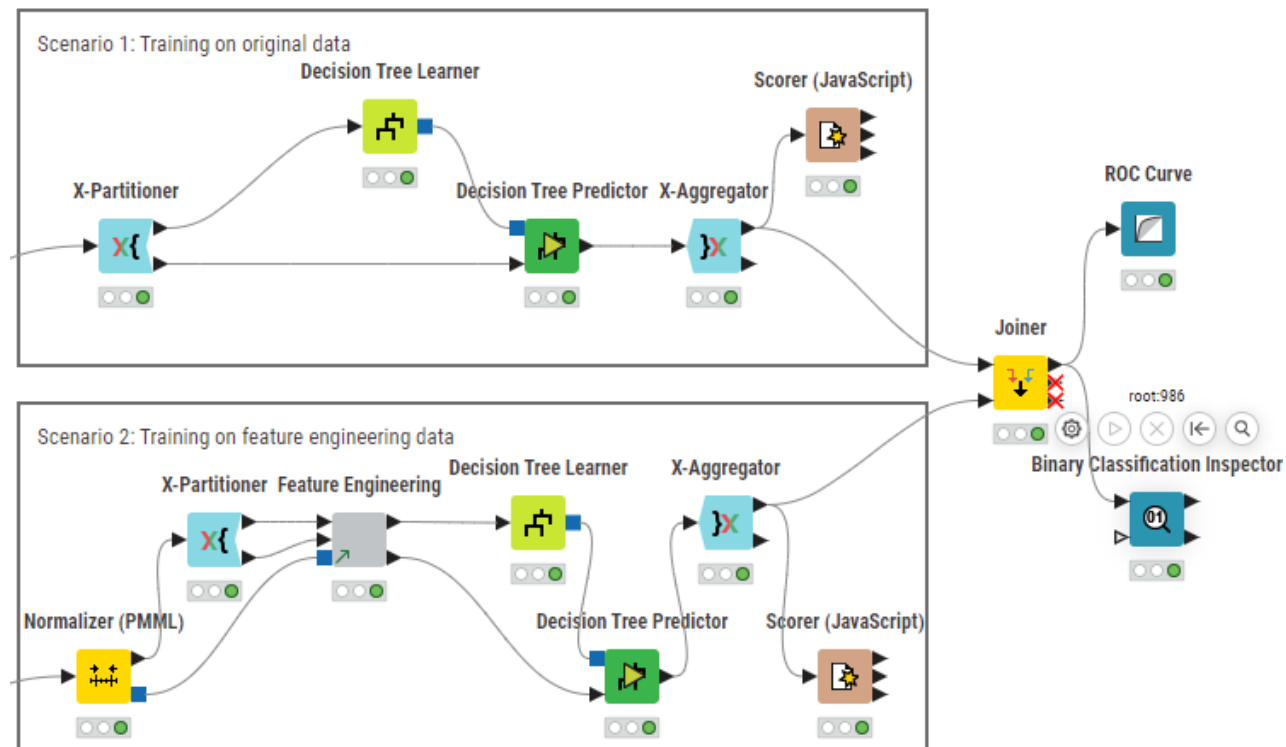
Feature Engineering Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	324	7098	2563	15	95.58%	4.37%	95.58%	26.53%	8.35%
Normal	2563	15	324	7098	26.53%	99.42%	26.53%	95.58%	41.88%

ROC Curve: Naïve Bayes

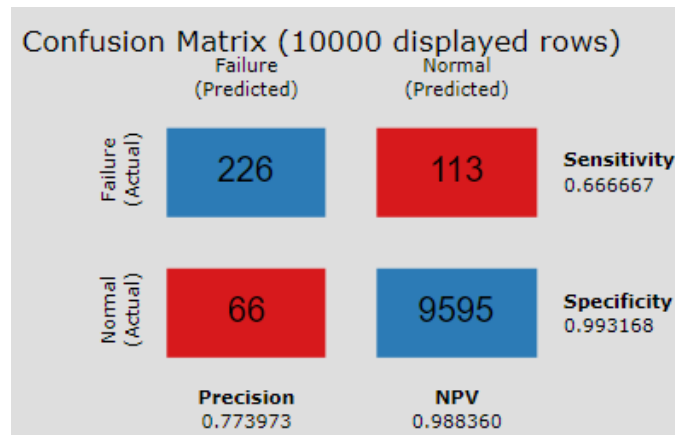


Classifier: Decision Tree

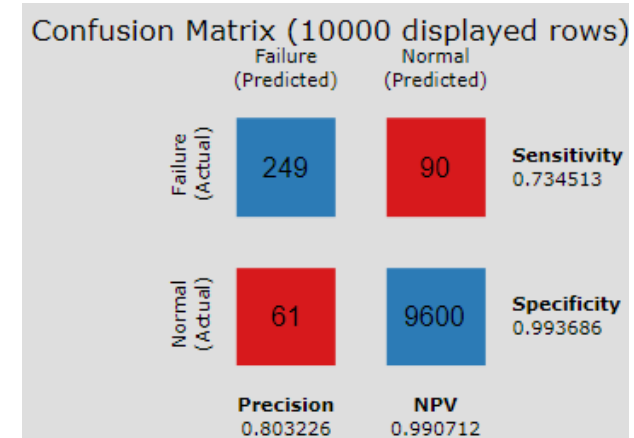


Confusion Matrix: Decision Tree

Original Data



Feature Engineering Data



Performance Metrics: Decision Tree

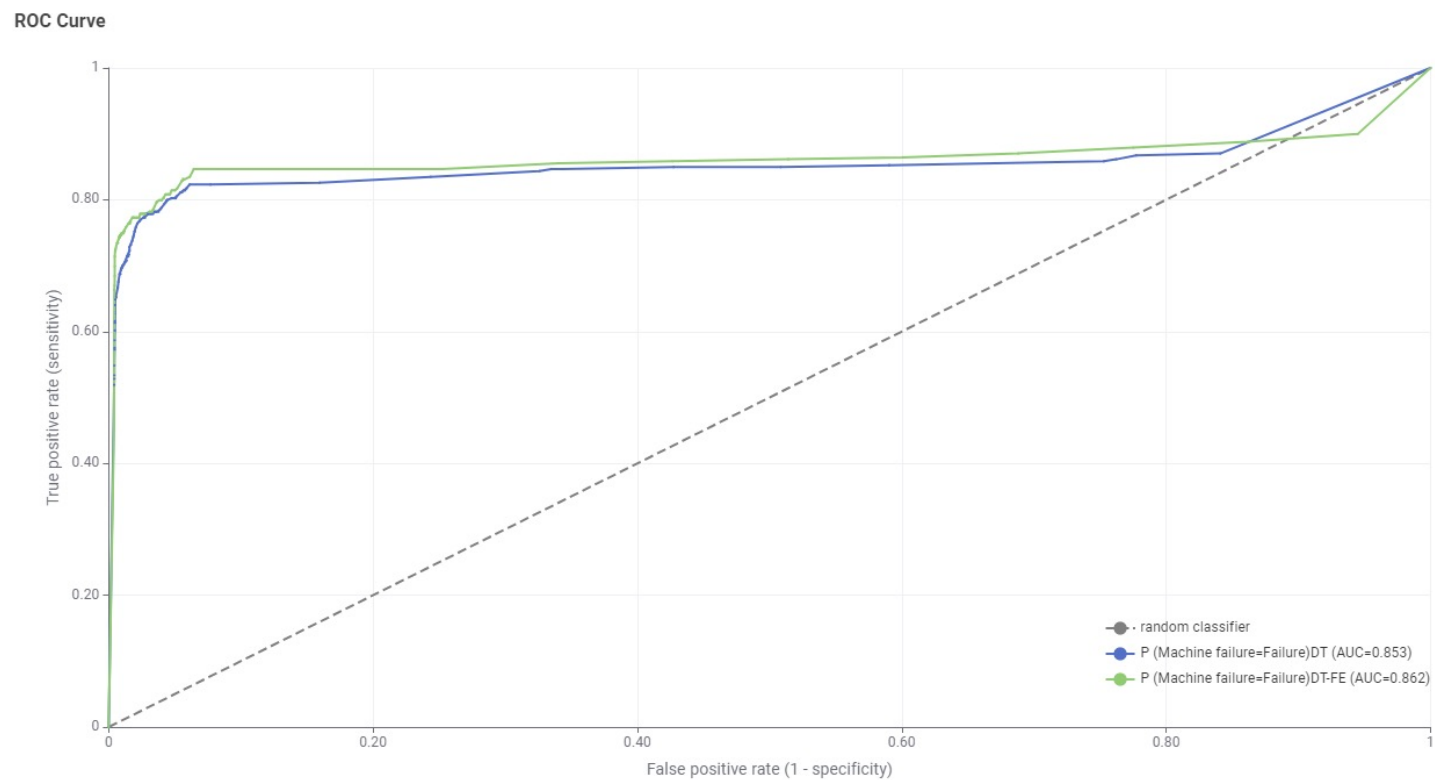
Original Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	226	66	9595	113	66.67%	77.40%	66.67%	99.32%	71.63%
Normal	9595	113	226	66	99.32%	98.84%	99.32%	66.67%	99.08%

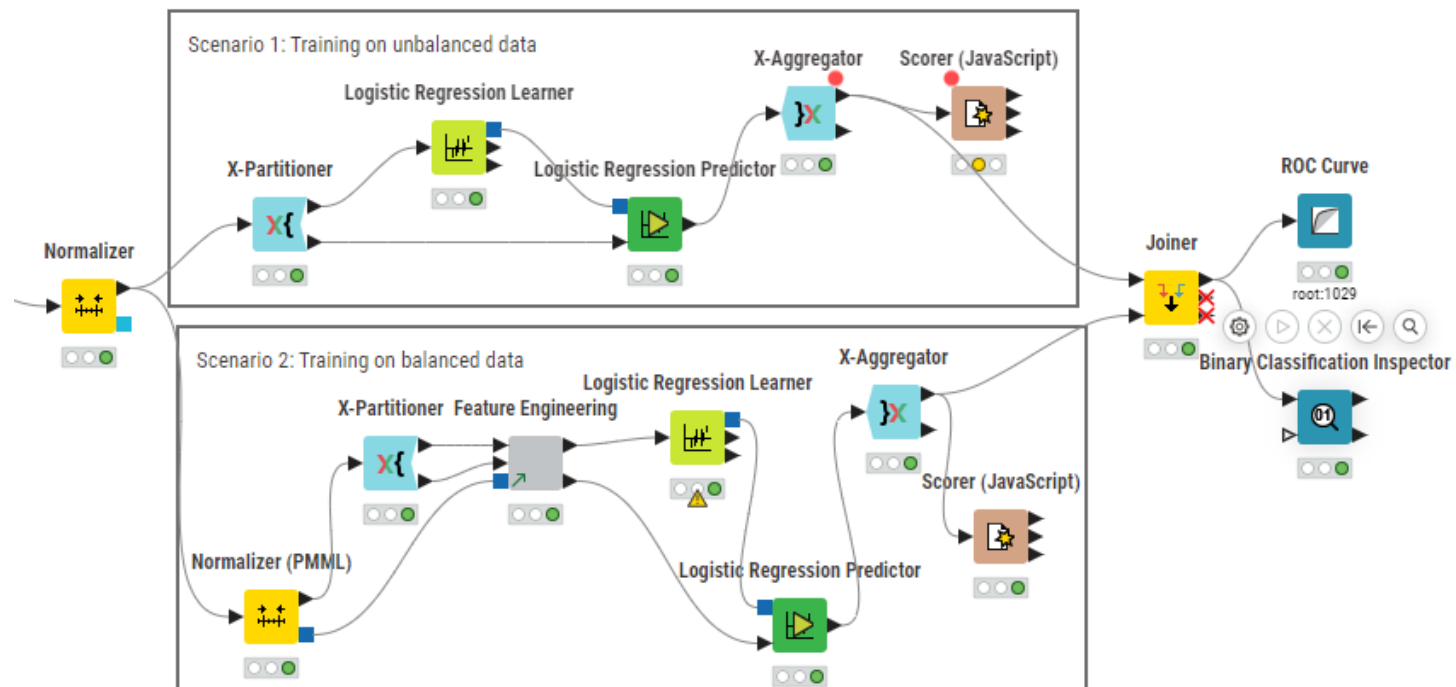
Feature Engineering Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	249	61	9600	90	73.45%	80.32%	73.45%	99.37%	76.73%
Normal	9600	90	249	61	99.37%	99.07%	99.37%	73.45%	99.22%

ROC Curve: Decision Tree

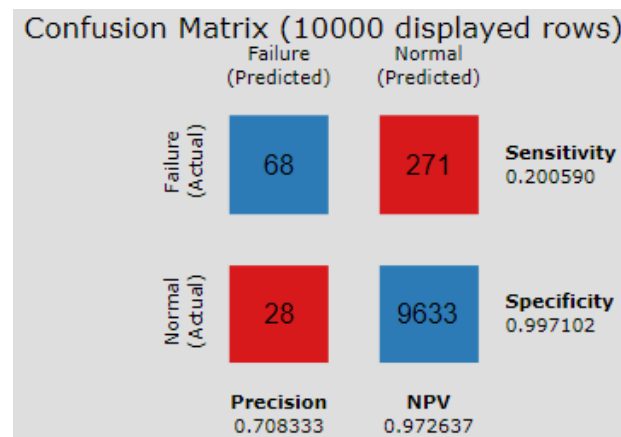


Classifier: Logistic Regression

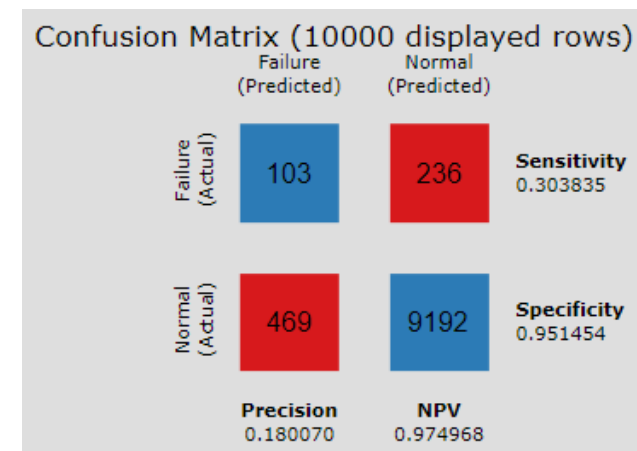


Confusion Matrix: Logistic Regression

Original Data



Feature Engineering Data



Performance Metrics: Logistic Regression

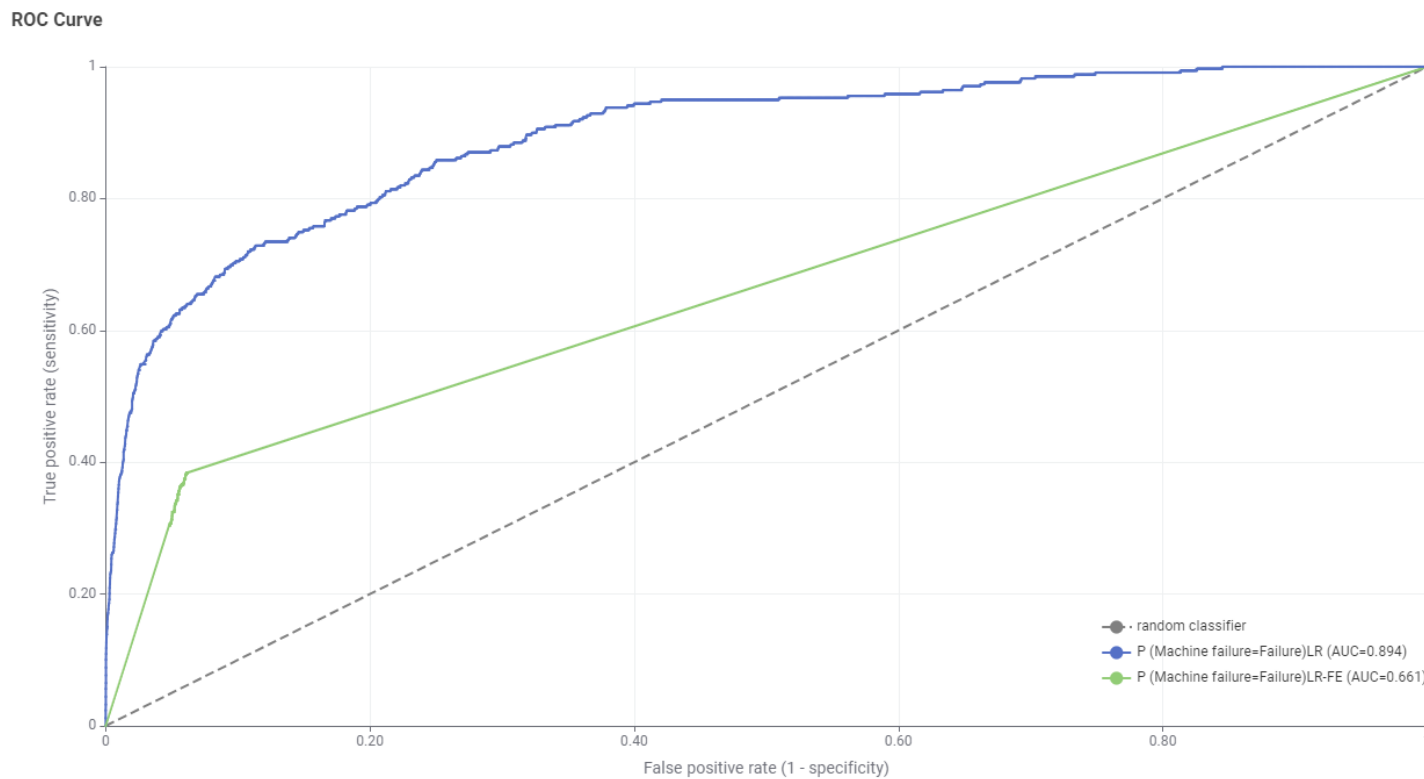
Original Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	69	28	9633	270	20.35%	71.13%	20.35%	99.71%	31.65%
Normal	9633	270	69	28	99.71%	97.27%	99.71%	20.35%	98.48%

Feature Engineering Data

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	104	469	9192	235	30.68%	18.15%	30.68%	95.15%	22.81%
Normal	9192	235	104	469	95.15%	97.51%	95.15%	30.68%	96.31%

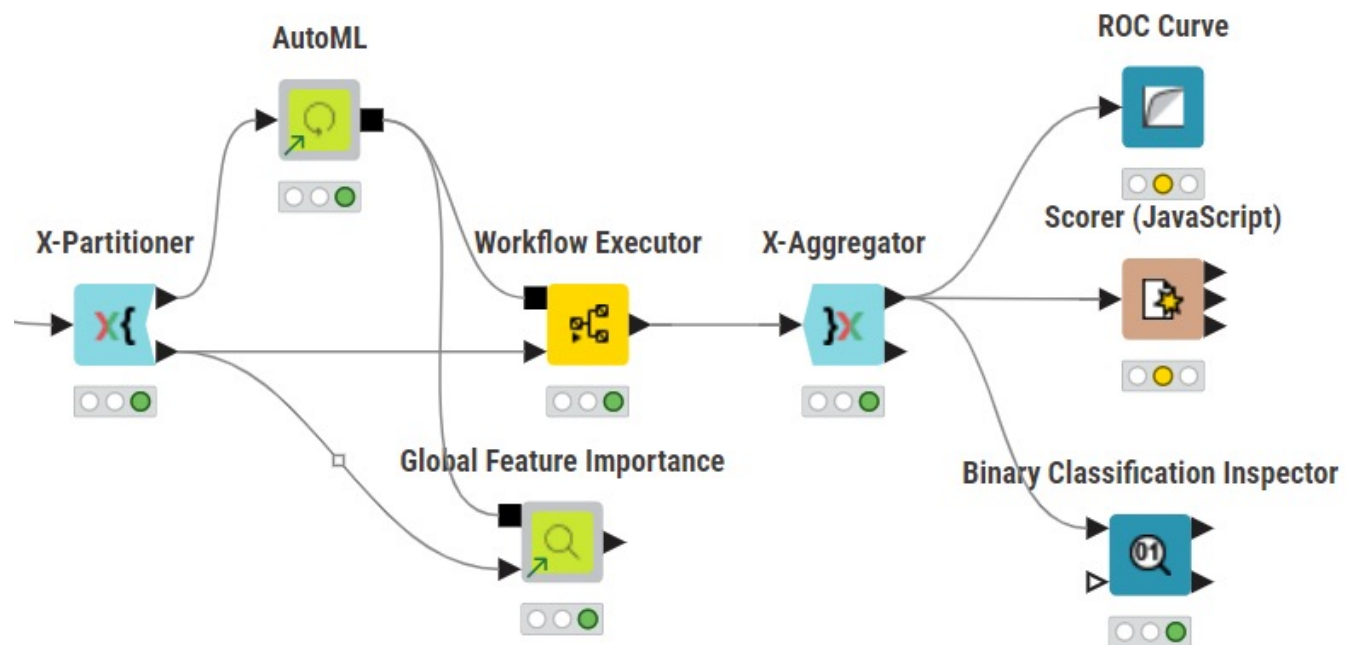
ROC Curve: Logistic Regression



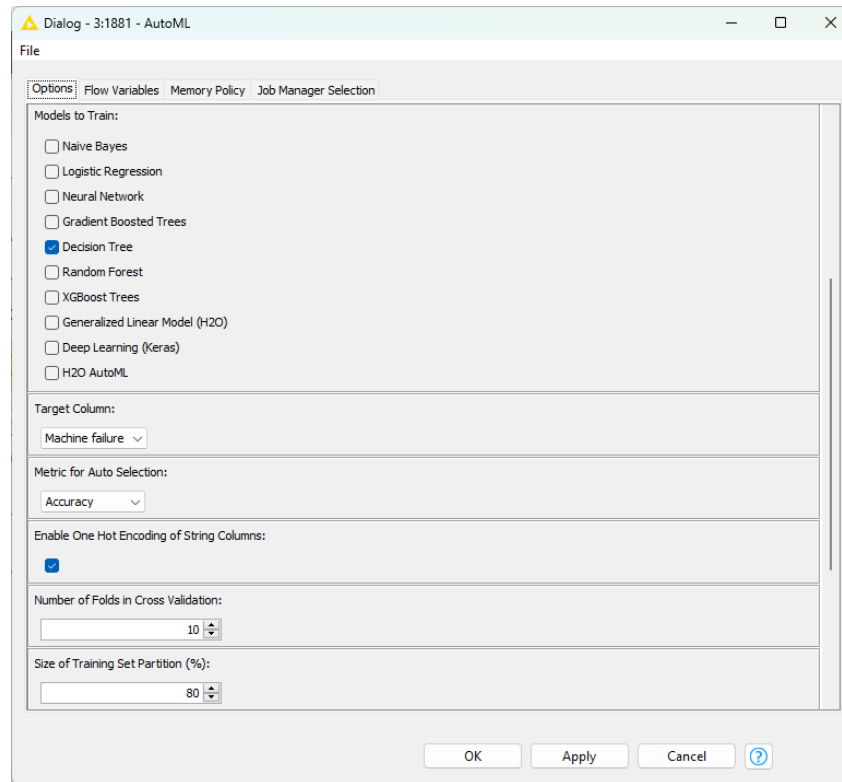
KNIME: AutoML



KNIME: AutoML



Important Setting: AutoML



The screenshot shows the 'Dialog - 3:1881 - AutoML' window. It has a 'File' menu and four tabs: 'Options' (selected), 'Flow Variables', 'Memory Policy', and 'Job Manager Selection'. The 'Options' tab contains the following settings:

- Models to Train:** A list of model types with checkboxes. 'Decision Tree' is selected (checked), while others are unchecked.
- Target Column:** A dropdown menu set to 'Machine failure'.
- Metric for Auto Selection:** A dropdown menu set to 'Accuracy'.
- Enable One Hot Encoding of String Columns:** A checkbox that is checked.
- Number of Folds in Cross Validation:** A spinner box set to 10.
- Size of Training Set Partition (%):** A spinner box set to 80.

At the bottom of the dialog are four buttons: 'OK', 'Apply', 'Cancel', and a help icon (question mark in a circle).

AutoML: Classification Metrics

Scorer View

Confusion Matrix



	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	221	118	65.19%
Normal (Actual)	55	9606	99.43%
	80.07%	98.79%	

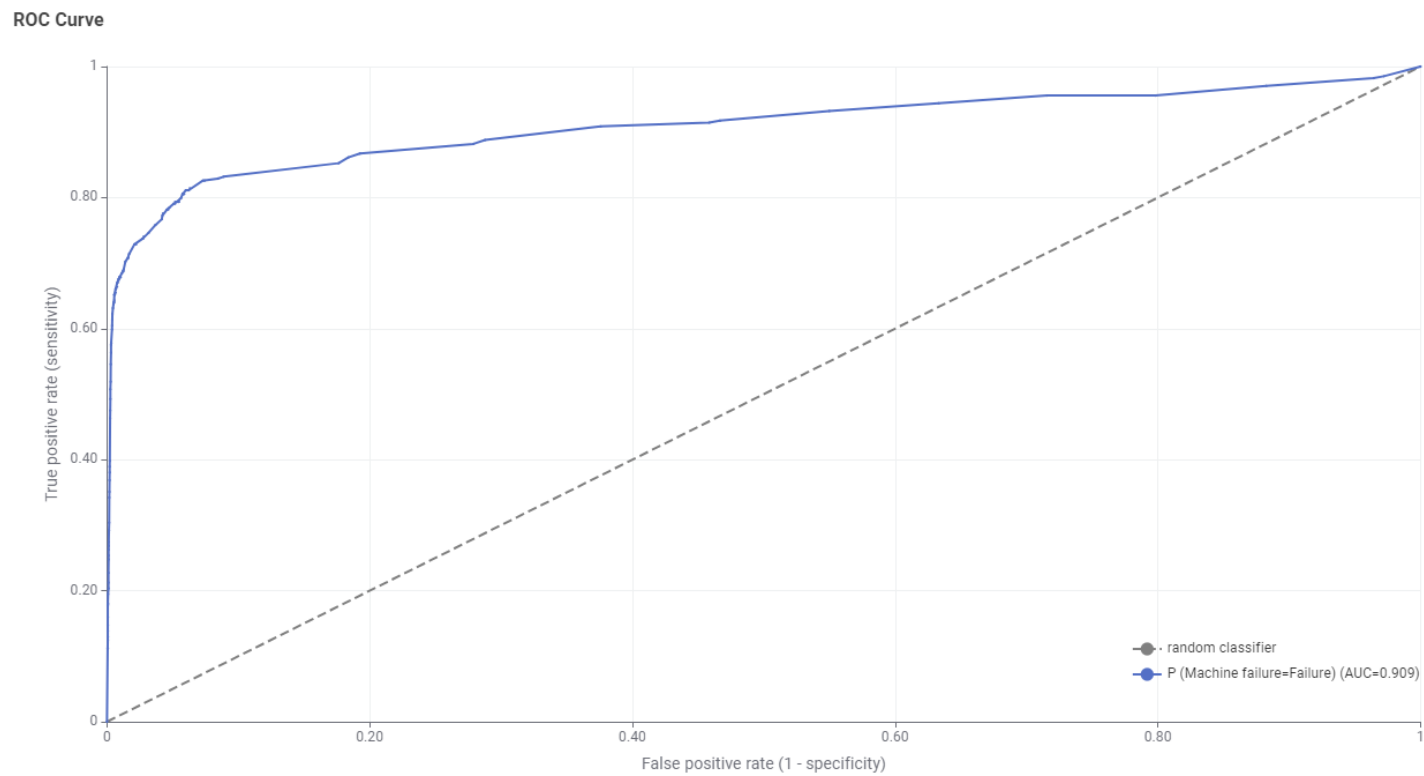
Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	221	55	9606	118	65.19%	80.07%	65.19%	99.43%	71.87%
Normal	9606	118	221	55	99.43%	98.79%	99.43%	65.19%	99.11%

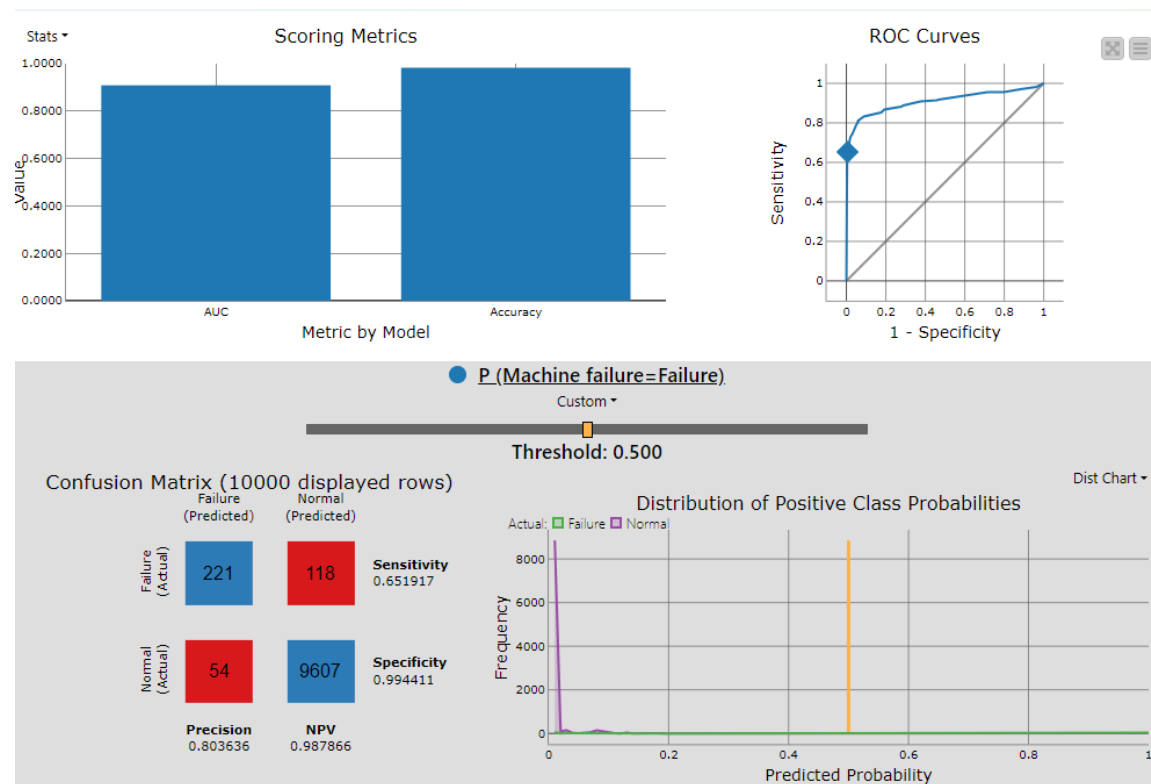
Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
98.27%	1.73%	0.710	9827	173

AutoML: ROC



Performance: AutoML



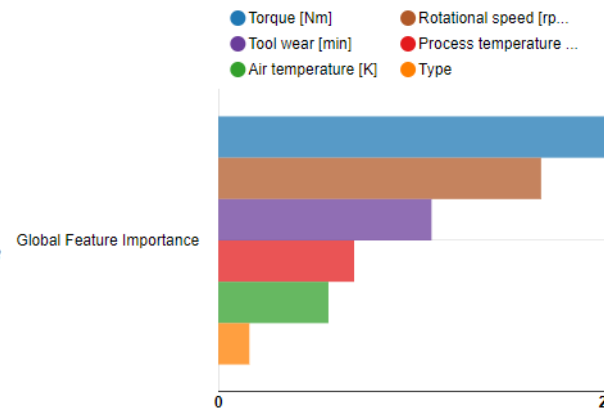
Feature Important

Surrogate Random Forest

A Global Surrogate Random Forest is a Random Forest model trained to approximate the predictions of the original model. Random Forest is trained on the standardly pre-processed input data with optimized parameters "Tree Depth", "Number of models" and "Minimum child node size".

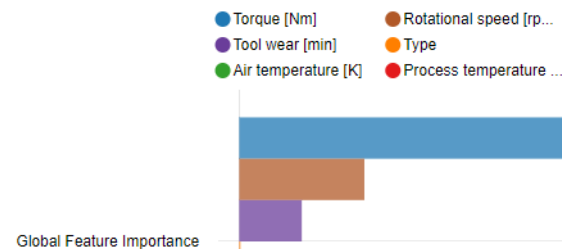
The surrogate model was trained successfully with *Accuracy* equal to 0.993 to the original model predicted class of interest "*Machine failure : Failure*".

Feature importance is calculated by counting how many times it has been selected for a split and at which rank (level) among all available features (candidates) in the trees of the random forest. Higher value indicates higher feature importance.



Permutation Feature Importance

Permutation feature importance measures the difference between the model performance score estimated on predictions using all the original features and the model performance score estimated on predictions using all the original features except one which was randomly permuted. If a feature was permuted several times, the average difference is calculated.



LAB: AutoML (Regression)

- <https://www.kaggle.com/datasets/yasserh/wine-quality-dataset/>
- Auto Model
- Feature Important